

The Heard Island Glaciology Program 2003-04

Studies on the morphology, dynamics, mass balance, and climatic setting of Brown Glacier.



Data report

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Contents

1	Introduction and background to the study	1
1.1	<i>Location and physiography of Heard Island.</i>	1
1.2	<i>Previous glaciological studies on Heard Island</i>	1
1.3	<i>Outline of the present study.</i>	2
1.4	<i>Data availability</i>	2
2	Brown Glacier	3
3	Survey Methods.	4
4	Ice velocity measurements	5
5	Resurvey of the glacier terminus.	7
6	Change in glacier surface elevation	7
7	Ice thickness	8
8	Digital elevation model.	10
9	Climate	12
9.1	<i>Automatic weather stations</i>	12
9.2	<i>Temperature.</i>	13
9.3	<i>Wind speed.</i>	14
9.4	<i>Solar radiation</i>	14
9.5	<i>Precipitation</i>	15
10	Ablation measurements.	15
11	Ice coring and crevasse sampling	16
12	Lagoon bathymetry surveys	16
13	Catchment runoff.	18
14	Accumulation and ablation indicators.	18
	References	19
	Appendices	
A1	Description of data files available on CD	20
A2	Digital elevation model	21
A3	Lagoon bathymetry	22
A4	Panorama photo record	26



The 2003-04 Heard Island glaciology team. Left to right: Doug Thost, Heather Kirkpatrick, Robb Clifton, Shavawn Donoghue, Martin Truffer.

Front Cover: Panoramic view from the cliffline east of Round Hill overlooking Brown Glacier and Brown Lagoon. Scarlet Hill is the prominent peak towards the centre of the photograph.

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STUDIES ON BROWN GLACIER 2003-04

1. Introduction and Background to the study

1.1 Location and Physiography of Heard Island

Subantarctic Heard Island is an ice-covered, mildly active composite shield volcano situated in the southern Indian Ocean at 53°05' S; 73°30' E, approximately 4350 km south-west of Western Australia, 4850 km southeast of southern Africa, and 1650 km north of the Antarctic continent. Together with the nearby McDonald Islands, and the Kerguelen Islands 433 km to the northwest, Heard Island is an emergent part of the extensive, predominantly igneous, submarine Kerguelen Plateau.

Heard Island extends 43 km from Cape Cartwright on Laurens Peninsula in the northwest to the end of Elephant Spit in the east (Fig. 1). The bulk of the island is approximately circular in plan with a 20 km diameter, and consists of the Big Ben volcano with its active conical vent (Mawson Peak, 2745 m) located on the south-western edge of a broad summit plateau (at c. 2400 m). The geologically younger Laurens Peninsula (Quilty & Wheller, 2000) is another distinct volcano, with Mt Dixon (706 m) at its summit.

Heard Island is spectacularly glacierised as a result of both its height and its location south of the Antarctic Convergence (or Polar Front), a discontinuity between cold Antarctic Surface Water and less cold Subantarctic Surface Water (e.g. Taljaard & van Loon, 1984). Glaciers cover an estimated 69% of the island (253 km² in 2000; Ruddell, in press, 2004). The majority of glaciers radiate from the summit region of Big Ben, with individual glaciers separated by well-developed buttresses (Quilty & Wheller, 2000). Many of the glaciers with westerly to south-westerly aspects are tidewater glaciers, whereas those with northerly and particularly north-easterly aspects are land terminating, or terminate in lagoons well inland from the ocean. Laurens Peninsula is also glaciated, albeit on a much smaller scale than Big Ben, with three separate, ice-covered areas.

1.2 Previous glaciological studies on Heard Island

Heard Island has been intermittently occupied since its discovery in 1833. After its early history as a venue for the exploitation of the local wildlife, it has been the site of infrequent visits by the scientific community. The longest period of continuous occupation was by the Australian National Antarctic Research Expeditions (ANARE), which established a base at Atlas Cove, and occupied the site from December 1947 until March 1955. With the exception of the only other wintering parties (1969 and 1992), all other occupations (a total of 15 including the 2003-04 expedition) were during summer.

Observations from 1947-55 suggested minor glacier recession was occurring, however it was the 1963 ANARE expedition that obtained the first evidence of major glacier retreat (Budd; 1964, 2000). Budd (2000) divides periods of observation and glacier response into an initial phase of

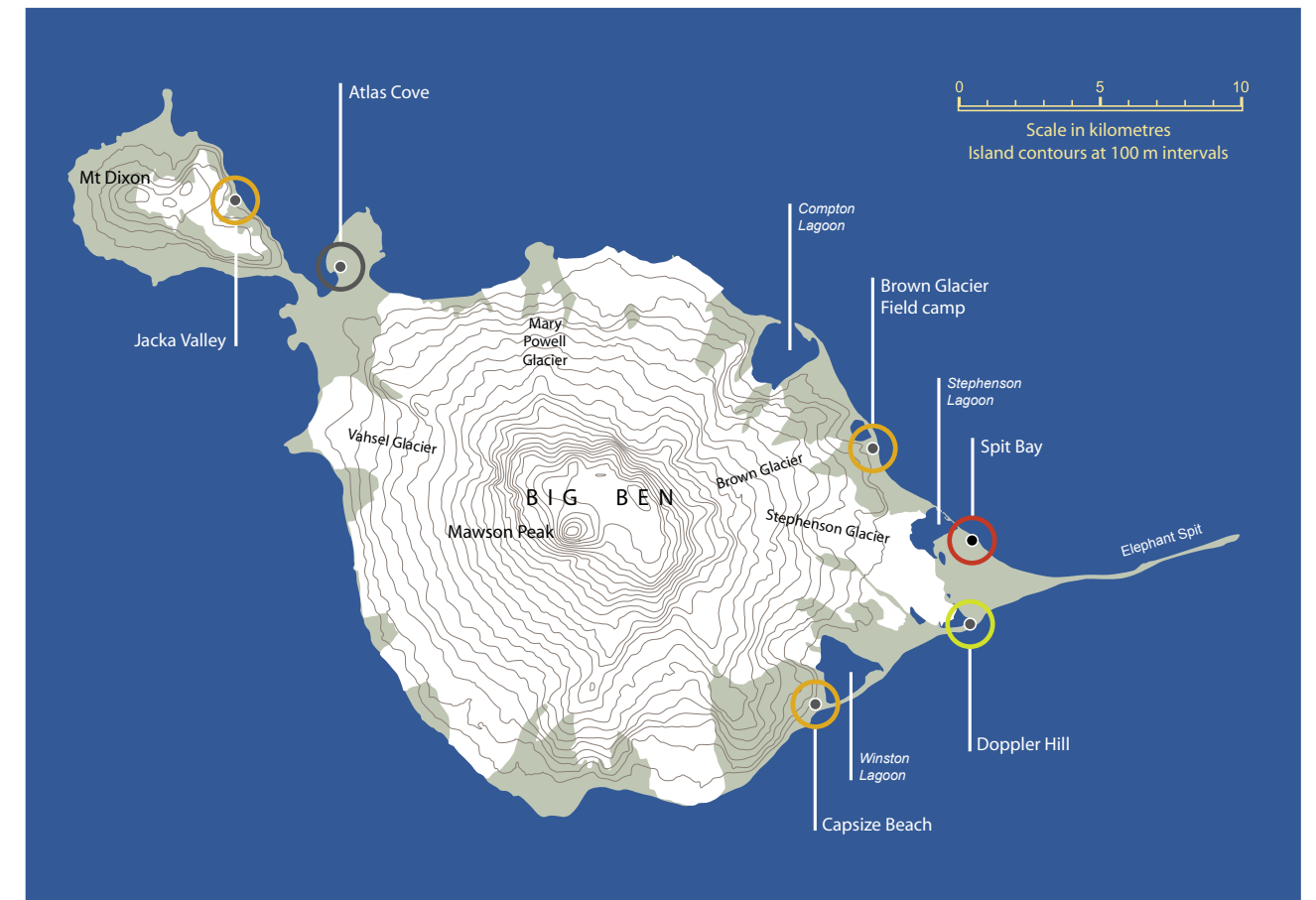


Figure 1. Map of Heard Island, with locations discussed in text. Modified from an original map by P. Boyer.

minimal change (1947-55), a subsequent period of rapid change (1963-71) in which a complex pattern of asynchronous glacier retreat and re-advance was observed, followed by continuing glacier recession (1983 – 2001). These fluctuations in glacier retreat and advance appear to be the result of changing air temperatures (and perhaps precipitation and cloud cover), and demonstrate sensitivity of the Heard Island glaciers to Southern Ocean climate change, as well as interactions between climate and glacier aspect, topography, and altitude range (Ruddell & Allison 1998; Budd 2000).

The main features of the Heard Island glaciers have been described by Lambeth (1951) and by Allison & Keage (1986). The first glacial modelling of Heard Island was by Budd & Allison (1975) with the application of an empirical scheme for estimating the dynamics of glaciers. Allison (1980) reported measurements of the velocity, strain-rate, surface profile, gravimetric ice thickness and ablation on the Vahsel Glacier in 1971, and presented an analysis of the mass balance of the glacier. Williams (1998) attributed the observed retreat of the Mary Powell and Vahsel Glaciers to climate change induced warming alone, with the modelled effects of increased geothermal heat flux from volcanic activity being negligible. Ruddell & Allison (1998) noted that

glacier retreat has typically been restricted to those glaciers that originate at elevations not much higher than 1500 m. Steady state analysis and time-dependent modelling of a number of glaciers by these authors suggested that a warming of at least 0.8 °C has occurred since 1947 (Ruddell & Allison 1998).

A reconnaissance study (ASAC project 1158) of the morphology, dynamics and mass balance of the Brown Glacier on the north-eastern side of Heard Island was undertaken between mid October and late November 2000 by Truffer *et al.* (2001). Using a portable radio echo sounder (RES) they found that most parts of the Brown Glacier are just over 100 m thick, with the greatest measured depth being 123 m. They estimated that since 1947 the glacier had retreated 1.1 km (decreasing in area by 33%) and had lost $1.6 \cdot 10^8 \text{ m}^3$ (38%) of its total volume, equivalent to an average surface lowering over the total area of 0.55 m a^{-1} . The equilibrium line of the Brown Glacier was estimated to be above 600 m elevation. Dust layers in the wall of a crevasse gave an estimate of annual net ice accumulation at the top of the glacier of 6.5 m a^{-1} , and annual net ablation at the snout was estimated to be nearly 7 m a^{-1} . The mass balance gradient was estimated at 0.017 a^{-1} , a high value typical of maritime climates. The maximum measured glacier velocity was 0.175 m d^{-1} ($\sim 64 \text{ m a}^{-1}$). There was no observed change in glacier speed during the 2000 field season.

Truffer *et al.* (2001) noted that there are significant climatic variations over Heard Island, with the eastern end of the island and the Brown Glacier region in particular having warmer summer temperatures than the western end. The warmer temperatures are due to the frequent occurrence of föhn winds, which may persist for several hours at a time. During these events, wind speed is usually very high ($>25 \text{ m s}^{-1}$) and air temperature over the glacier surface can increase to above $+10 \text{ °C}$. These events have a significant impact on glacier ablation. Truffer *et al.* (2001) conclude that Brown Glacier is expected to continue retreating.

1.3 Outline of the present study

This study is a continuation of ASAC 1158, under ASAC 2363. Two of the party involved in the 2000 field season (Martin Truffer and Doug Thost) returned to the island, along with PhD student Shavawn Donoghue. The party was complemented with additional field support from Heather Kirkpatrick (expedition Field Training Officer) and Robb Clifton (expedition Field Leader).

The objectives for the field study were to:

- Resurvey the glacier surface and terminus using differential GPS (DGPS), to measure changes that had occurred in the three years since the initial survey;
- Establish and resurvey a grid of markers similar to, but more extensive than, the grid surveyed during 2000 to better define flow vectors and velocities;
- Determine ice thickness along a number of profiles using RES, to supplement the single detailed profile obtained during 2000, allowing better estimates of ice-flux at a number of elevations;

- Deploy three weather stations at low, intermediate, and high elevations to characterise micro-climatic variations over the glacier surface, and help refine radiation models for the glacier;
- Collect a number of 10 m ice cores using a hand-auger for detailed oxygen isotope ratio ($\delta^{18}\text{O}$) and ion chemistry analysis, complemented by samples from deeper within crevasses, to allow determination of accumulation rates;
- Deploy two sonic rangars to measure variations in surface ablation and accumulation over the field season;
- Resurvey the bathymetry of Brown Lagoon using a depth sounder, and if possible, survey Stephenson, Compton, and Winston Lagoons, to determine lake volume;
- Measure the outflow from Brown Lagoon over the summer as a gross indication of ablation rate.

1.4 Data Availability

This report presents an outline of the work undertaken during the 2003-04 field season, and provides a reference for researchers who may return to the island in future years. More detailed interpretations of the results will be presented elsewhere.

The data collected during the 2003-04 field season is compiled on the accompanying CD, and is listed in Appendix A1. The CD also contains high resolution files of figures used in the text of this report, and should be referred to if clarity is an issue on printed versions, particularly if the report is reproduced on A4-size paper. The data is also archived with the Australian Antarctic Division, Australian Antarctic Data Centre, and will be available from their website:

<http://www-aadc.aad.gov.au>

2. Brown Glacier

Brown Glacier flows from an altitude of about 1100 m to 80 m a.s.l. on the north-eastern side of the island (Figs. 1, 4). It is presently ~4.8 km long. An aerial photograph taken in 1947 (Fig. 2) shows it and the nearby Stephenson Glacier to have been active tidewater glaciers with terminal ice cliffs at least 15 m high. By March 1963 Brown Glacier had retreated more than 100 m inland, leaving a small lagoon separated from the sea by a boulder beach (Budd, 1964, 2000), and by 1980 a proglacial lake had developed. The glacier eventually retreated past this lake (currently 0.416 km² with a maximum recorded depth of -56 m) and became land terminating circa 1985 (Fig. 3). Between 1947 and 2003, Brown Glacier has retreated by a total of 1.17 km ($1\sigma = 80$ m) at an average rate of 20.9 m a⁻¹.

Figure 2. *Aerial photograph from the Walrus flight of 1947, showing Brown Glacier as a healthy tidewater glacier with a 15 - 20 m high terminating ice-cliff. From RAAF photo 616, © Commonwealth of Australia, Australian Antarctic Division.*



Figure 3. *Brown Lagoon and Brown Glacier, December 12th 1987. The glacier has retreated from the lagoon and become land terminating. Photo F43/35 © Jenny Scott.*

Figure 4. *Brown Lagoon and Brown Glacier, December 30th 2003. Photo DT11_0080.jpg.*



3. Survey Methods

All the surveying in the 2003-04 field season was done using GPS methods. The GPS base station established in 2000 was reoccupied. It is located on an isolated rock south of the field camp and is marked with a chiselled and painted cross and a rock cairn. *Leica Geosystems* and *Trimble* receivers were used in the field and data was reduced using the manufacturers' proprietary software. The position of the Brown Glacier local base station is 53° 04' 50.6748" South, 73° 40' 28.3220" East, 68.1100 m (above ellipsoid height).

Twenty-six 2.4 m long fibreglass poles were deployed on the surface of the glacier. Two were lost after they melted out and were subsequently reburied by snow (at sites BG051 and BG151; Fig. 5 and Table 1). Each of the poles was visited several times throughout the 9 week field season, to measure the amount of snow/ice melt, and to resurvey the position using DGPS to determine the ice velocity at that location.

Additional surveys included kinematic DGPS surveys of the glacier terminus, the lagoon shoreline, and a resurvey of a track completed in the 2000 season to determine altitudinal change in the ice surface. All data were reduced to determine the position (latitude, longitude, ellipsoid height) in the WGS84 system. The elevation above sea level was obtained by subtracting 40 m from the ellipsoid height; the same procedure has been used in the creation of the new digital elevation model (DEM).

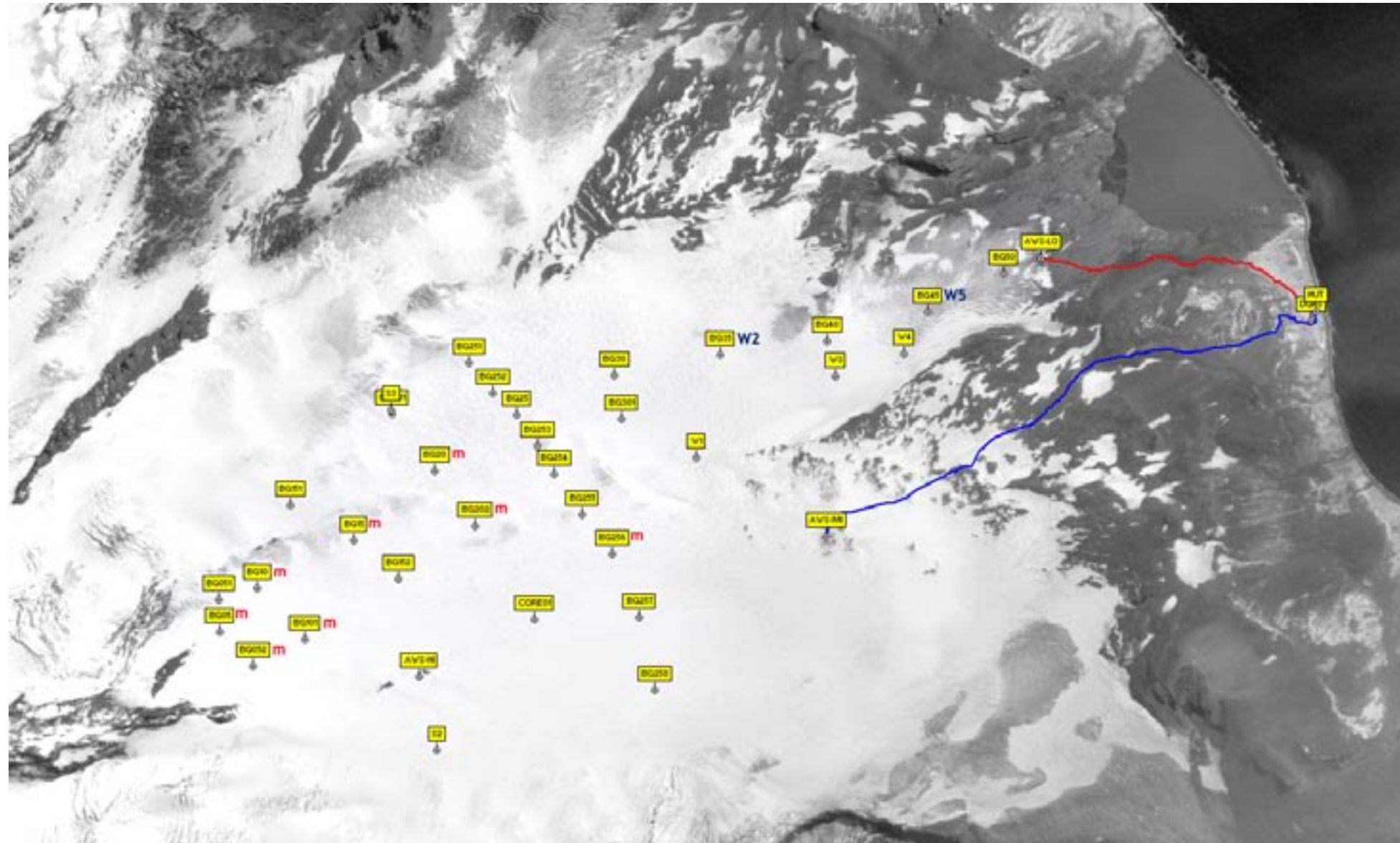


Table 1. Listing of all survey sites, sample sites, and weather stations (see Fig. 5). Asterix denotes lost survey cane. (m) denotes magnet left in situ on the 13th February. Wn denotes ablation wire left in situ 4th-6th February. Chemistry samples relate to the following locations on Fig. 5: BC1 at S3; BC2 at BG152; BG-1 at CORE01; SC1 and SG-1 at S2. Coordinates are UTM zone 43.

Locality	Easting	Northing	Height a.s.l. (m)	Locality	Easting	Northing	Height a.s.l. (m)
AWS-HI	407039	4116152	920	BG254	407663	4117140	682
AWS-LO	409940	4118215	115	BG255	407799	4116945	703
AWS-MI	408947	4116851	550	BG256 (m)	407944	4116754	705
BG05 (m)	406098	4116353	1030	BG257	408073	4116450	715
BG051*	406092	4116509	1000	BG258	408152	4116097	715
BG052 (m)	406258	4116192	1034	BG30	407940	4117620	582
BG10 (m)	406272	4116569	955	BG301	407979	4117408	580
BG101 (m)	406500	4116324	972	BG35 (W2)	408439	4117729	492
BG15 (m)	406727	4116806	866	BG40	408943	4117800	407
BG151*	406426	4116972	851	BG45 (W5)	409414	4117945	296
BG152 crevasse 3	406935	4116622	871	BG50	409769	4118135	179
BG20 (m)	407100	4117146	752	CORE01	407579	4116432	799
BG201	406897	4117419	756	S2 crevasse 1	407127	4115797	922
BG202 (m)	407294	4116884	779	S3 crevasse 2	406894	4117439	756
BG25	407482	4117423	673	W1	408331	4117225	545
BG251	407257	4117673	682	W3	408984	4117630	405
BG252	407373	4117525	684	W4	409301	4117742	341
BG253	407582	4117276	678				

Table 2. Time between surveys for both field seasons. Epoch 1 (2003-04) was over such an extended period due to repeated failure of the DGPS equipment.

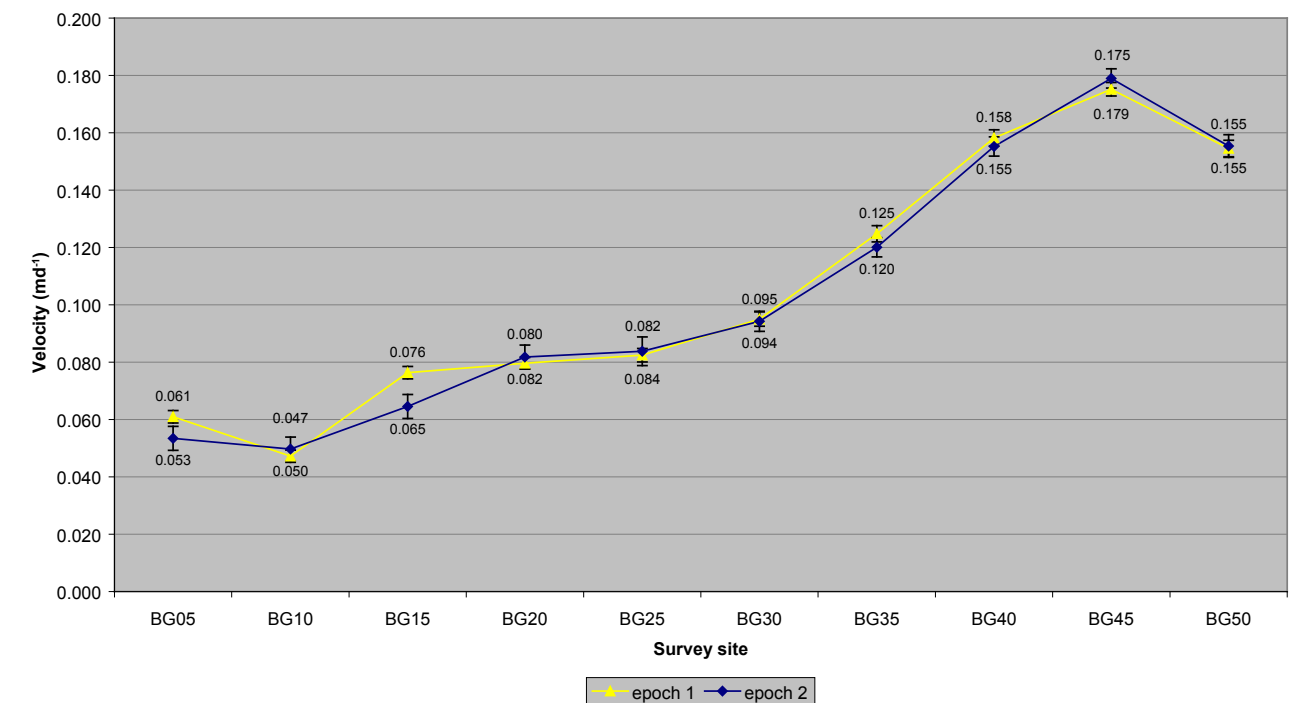
Season 2000	Average number of days between measurements
epoch 1: 27-28 OCT to 7-9 NOV	8.9
epoch 2: 7-9 NOV to 21-22 NOV	10.1
Season 2003-04	
epoch 1: 22-28 DEC 2003 to 20-26 JAN 2004	30.5
epoch 2: 20-26 JAN to 9-12 FEB	16.8

4. Ice velocity measurements

Average surface ice velocity (Figs 6-9, in m.d^{-1}) at each surface survey site was determined over two epochs by measuring the difference in position of the survey marker using DGPS divided by the number of days between surveys (Table 2). Greatest velocities were measured early in the season, with a maximum of 0.168 m.d^{-1} at BG45 (0.177 m.d^{-1} in 2000) on the lower and steeper part of the glacier. Surface ice velocities decreased by as much as 36% over the two epochs (at BG25 from 0.110 to 0.071 m.d^{-1}) with an average decrease for all sites of 22% (0.101 to 0.069 m.d^{-1}).

The higher and lower velocities recorded at BG15 and BG05 (respectively) during 2004 compared with the 2000 survey are inconsistent with results from the other sites. With the exception of sites BG05 and BG15, the location of all survey sites between the two field seasons are coincident or very nearly so. Site BG15 (2004) was located 91 m south-east of BG15 (2000), closer to the central flow line of the glacier, which may explain the higher velocities recorded there. BG05 (2004) was located 80 m north north-east of and down slope from BG05 (2000). The surface gradient was less at the 2004 site which may explain the lower velocities.

Figure 6. Longitudinal velocity profile for the main survey sites, November 2000.



It is notable that the higher sites (BG05 - BG15) only show a modest reduction in velocity with time compared with all the sites down slope. A possible explanation may be that the higher sites, being colder, do not receive as much melt water at the base of the glacier, and are less lubricated. It is equally notable that velocities at the lower sites have slowed. February and March are still warm months on Heard Island (average temperatures measured by the Bureau of Meteorology data buoy at Spit Bay 2003-04: Dec 3.5; Jan 5.1; Feb 4.2; Mar 5.3; Apr 3.8), so it is to be expected that surface melting would continue to provide melt water to the glacier base.

It is possible that higher velocities than were recorded in November 2000 at BG40, BG45 and BG50 may have been averaged lower in epoch 1 (2003-04) due to the 30 day period between measurements. This applies to all sites, but is most obvious for the three lower sites where the epoch 1 (2003-04) velocity is less than the velocity for both epochs in 2000.

Figure 7. Longitudinal velocity profile for the main survey sites, 2003-04 field season. The average velocity for the November 2000 field season is also shown.

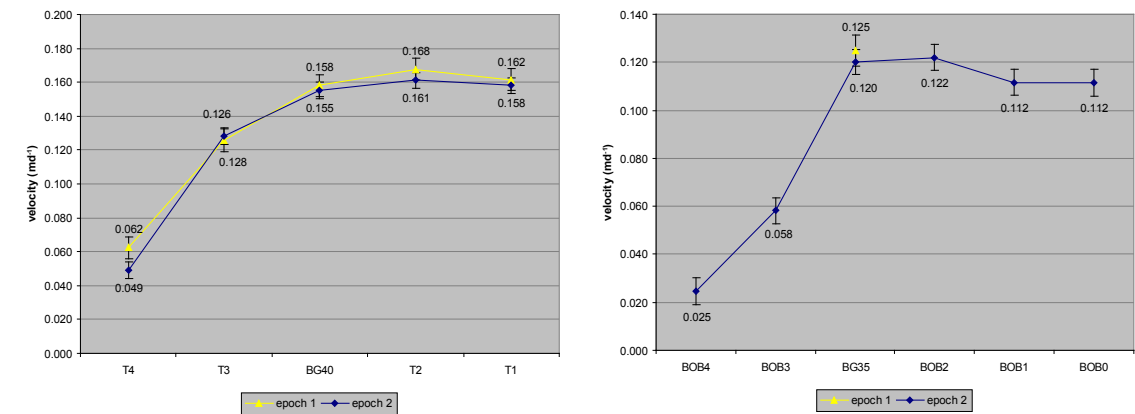
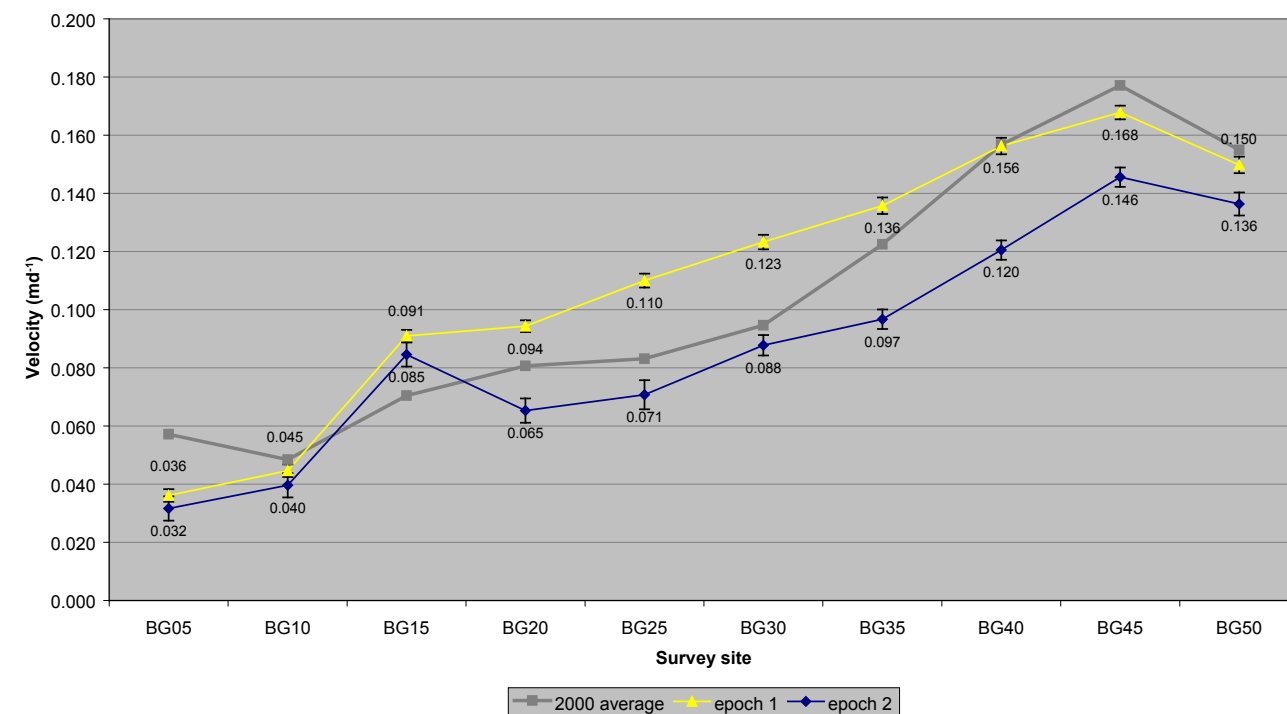


Figure 8. Transverse velocities for profiles across BG40 (left) and BG35 (right) during the 2000 field season.

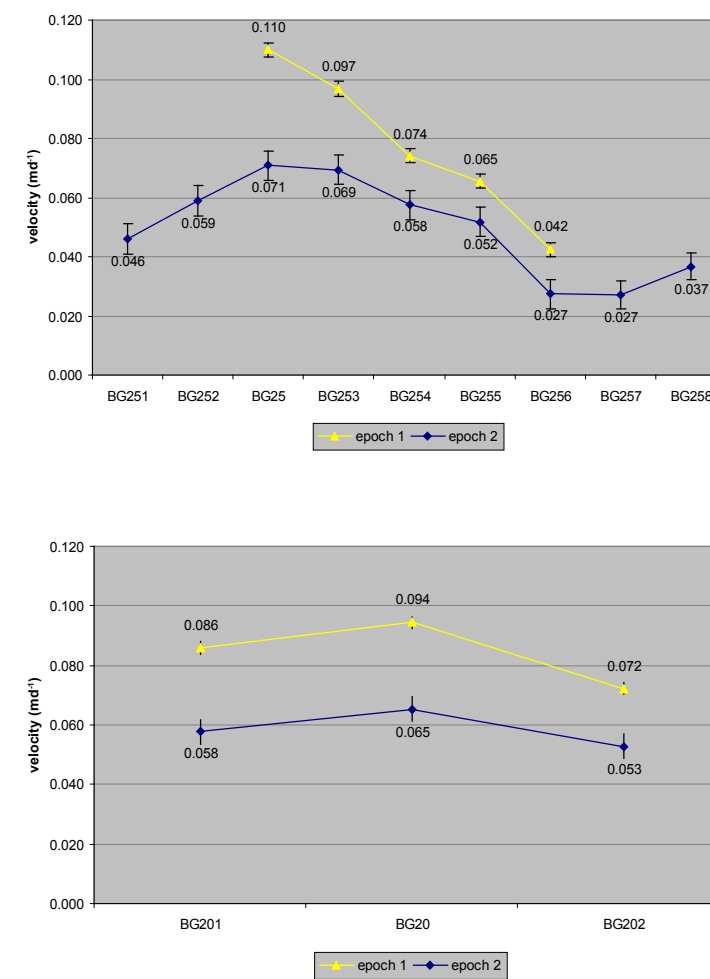


Figure 9. Transverse velocities for profiles across BG25 (upper) and BG20 (lower) during the 2003-04 field season.

5. Resurvey of the glacier terminus

A comparison between the terminus surveys undertaken on the 23 November 2000 and the 17 February 2004 shows a marked retreat in the terminus position (Fig. 10). The November 2000 survey was hampered by snow obscuring the ice-rock boundary, with positional errors for the boundary estimated at 20 m. Some snow ramps were still present in December 2003, but the boundary was generally well defined. These problems were not apparent during the February 2004 survey. The average retreat for the East terminus position, measured in the approximate direction of ice flow (between the November 2000 and the December 2003 surveys), is 63.0 m, equating to 20.3 m a^{-1} . This figure is comparable to the average retreat rate from 1947 to 2003 of 20.9 m a^{-1} , as well as for between 1980 and 2003 (20.6 m a^{-1}). Other values are presented in Table 3. Note the very high calculated retreat rate for the East terminus area between December 2003 and February 2004. During this period the lower glacier became increasingly crevassed (and ultimately almost impassable), leading to a dramatic increase in the surface area exposed to high summer air temperatures.

6. Change in glacier surface elevation

A resurvey of the kinematic profiles on the glacier surface undertaken in 2000 was completed in 2003-04 (Figs 11-13). A track approximating the 2000 survey was followed using a hand-held *Garmin* GPS V, traversed in such a way that the earlier track was intersected multiple times. In this way a total of 684 crossing points were obtained. The 2003-04 profiles were corrected to the beginning of the field season (late December 2003), using the nearest of 19 ablation markers that covered the glacier. The 2000 profiles were collected in November. Towards the end of that month most sites on the glacier were still accumulating snow. Other personnel revisited the glacier in mid January 2001 and reported melting of up to 1 m at the lower sites. It was assumed that the additional accumulation until late December roughly balanced this ablation and that the profile to profile comparison is from December 2000 to December 2003, thus removing seasonal effects.

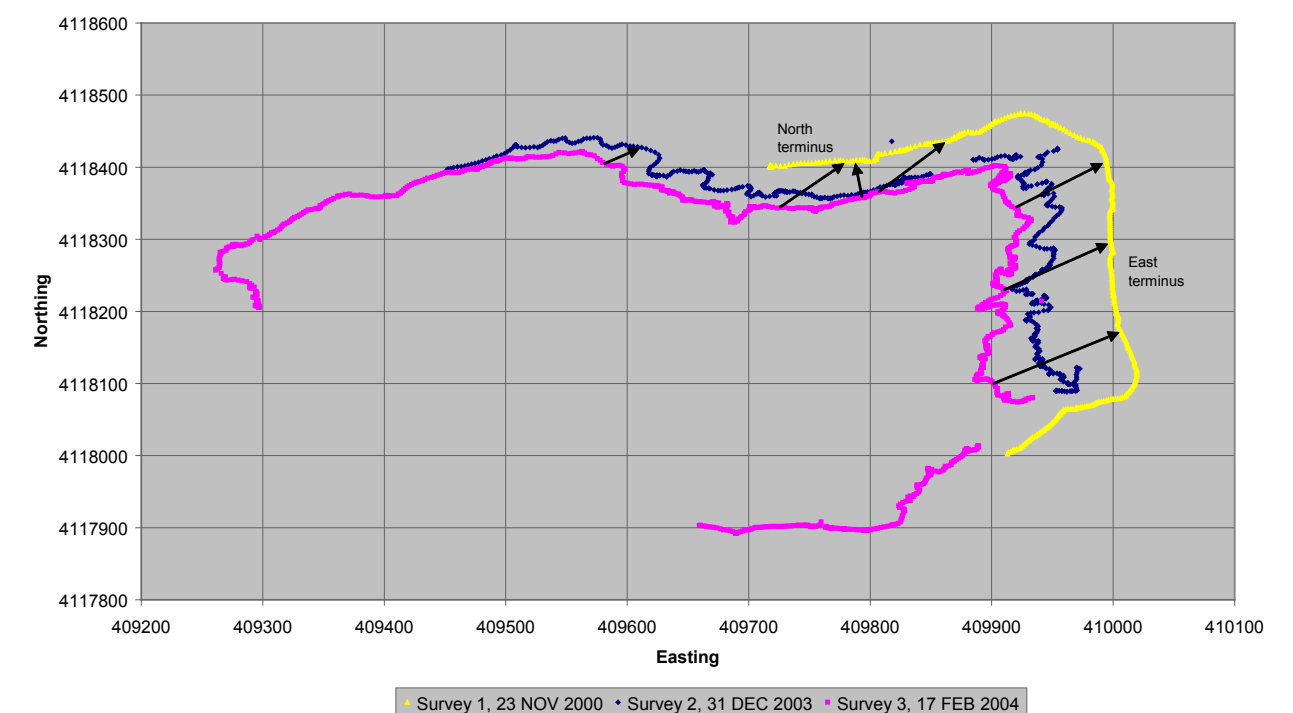
The elevation difference at the crossing points was used to calculate the total volume change. The glacier was divided into a regular grid, and a thickness change was calculated for each point as a distance-weighted average of all the crossing points within a 1000 m radius. The glacier surface has lowered by as much as 11 metres in the ablation area, and 4 metres in the accumulation area.

The between year profile-to-profile comparisons show a rate of ice loss of $8.0 \pm 1.3 \cdot 10^6 \text{ m}^3 \text{ a}^{-1}$. The error results from an estimated 0.4 m uncertainty in elevation due to correcting the profiles back to late December (see above), and an estimated 1 m error from the interpolation between the crossing points. The actual surveying errors are in the 1 cm range and therefore negligible. It is common that elevation changes are largest at lower elevations (e.g. Arendt *et al.*, 2002), but it is perhaps surprising that even the highest locations on the glacier are thinning significantly.

Table 3. Retreat rates for the glacier terminus. Average retreat distance (*retreat*) and standard deviation (*stdev*) in metres. The arrows in Fig. 10 show the measurement directions.

	1. Measured in direction of ice flow			2a. Measured in approximate direction of ice flow			2b. Measured normal to ice front		
	East terminus			North terminus			North terminus		
Epoch	retreat	stdev	m/a	retreat	stdev	m/a	retreat	stdev	m/a
Nov '00 to Dec '03	63.0	10.0	20.3				40.2	5.3	13
Nov '00 to Feb '04	99.7	13.3	30.8	65.9	6.1	20.4	47.1	6.6	14.6
Dec '03 to Feb '04	34.0	12.7	258.5	22.8	8.0	173.4			

Figure 10. Repeat DGPS surveys of the terminus position of Brown Glacier. The large arrows show the approximate direction of ice flow, and refer to measurement directions in Table 3. The small arrow on the "North terminus" is approximately normal to the ice front. The plan view is distorted due to the different scales used for Easting and Northing on this plot.



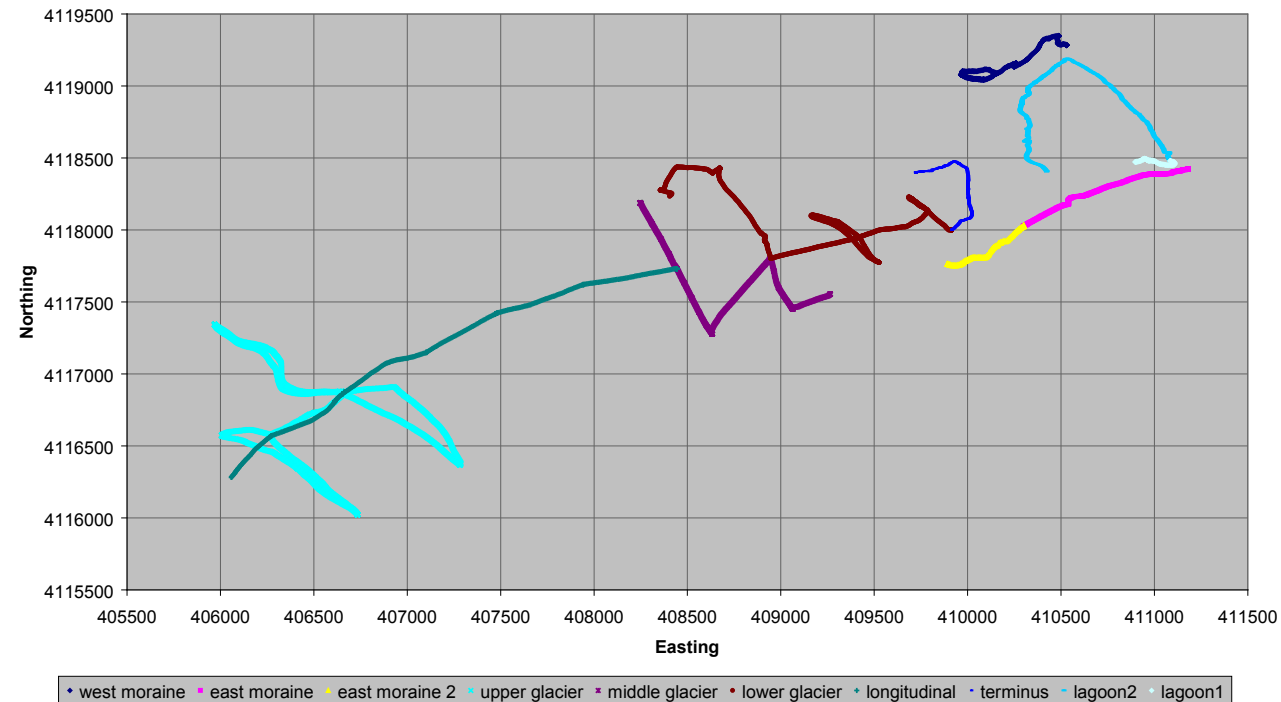
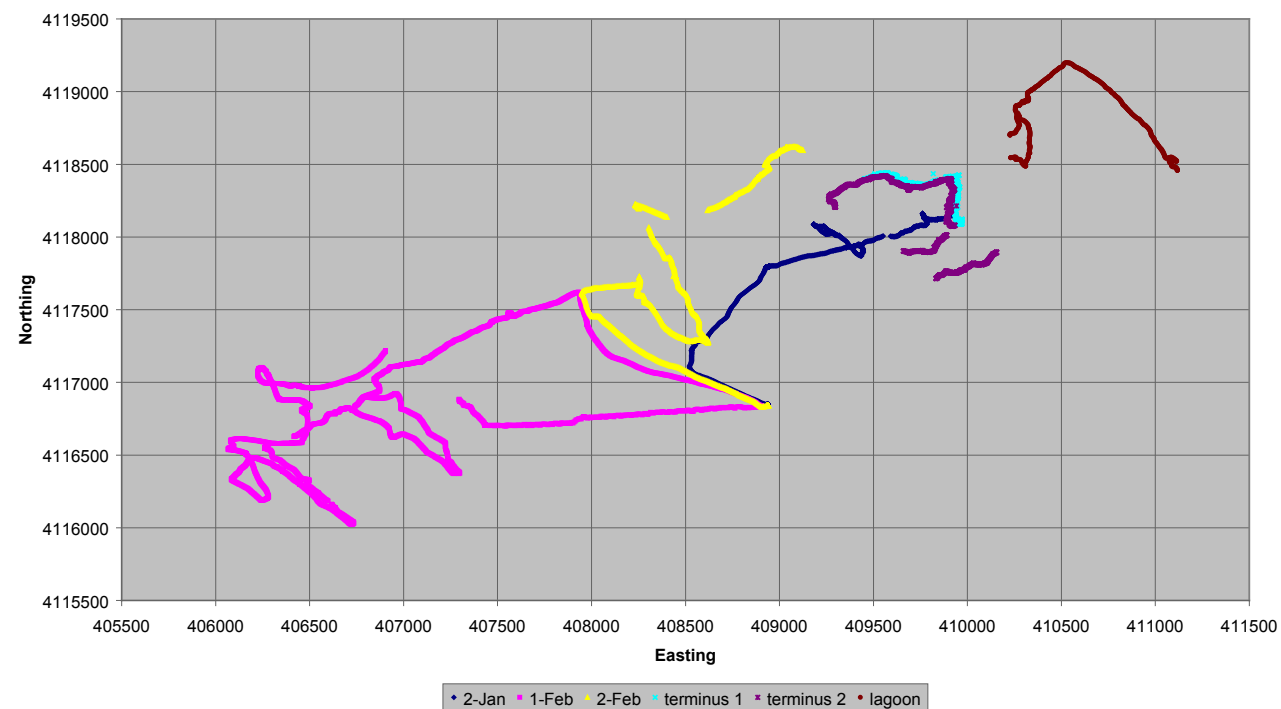


Figure 11. DGPS kinematic surveys on Brown Glacier during the 2000 field season. The plan view is distorted due to the different scales used for Easting and Northing on this plot.

Figure 12. DGPS kinematic surveys on Brown Glacier during the 2003-04 field season. The plan view is distorted due to the different scales used for Easting and Northing on this plot.



7. Ice Thickness

Ice thickness was measured with a portable low frequency RES system (Narod & Clarke, 1994) built by *Icefield Instruments Inc.* (Whitehorse, Canada). The length of the transmitting antennas determines the frequency of the wave. A 5 m dipole antenna was used, producing frequencies of 4.2 MHz.

A number of reflection points were acquired and lines tangential to the return ellipse define the glacier bed. Reflections are generally assumed to be within a plane perpendicular to the glacier surface, which is reasonable when the profiles are in a transverse direction in a simple valley glacier geometry.

One detailed profile at BG35 (Fig. 14) and several spot depths along the glacier were recorded in 2000. Two additional profiles were acquired this season on 28 January, at the BG25 and BG20 elevations (Fig. 15).

Figure 13. Martin Truffer carrying the Leica gear on a kinematic DGPS survey, 2 February 2004. Photo in the vicinity of BG40. Photo DT21_0082.jpg.



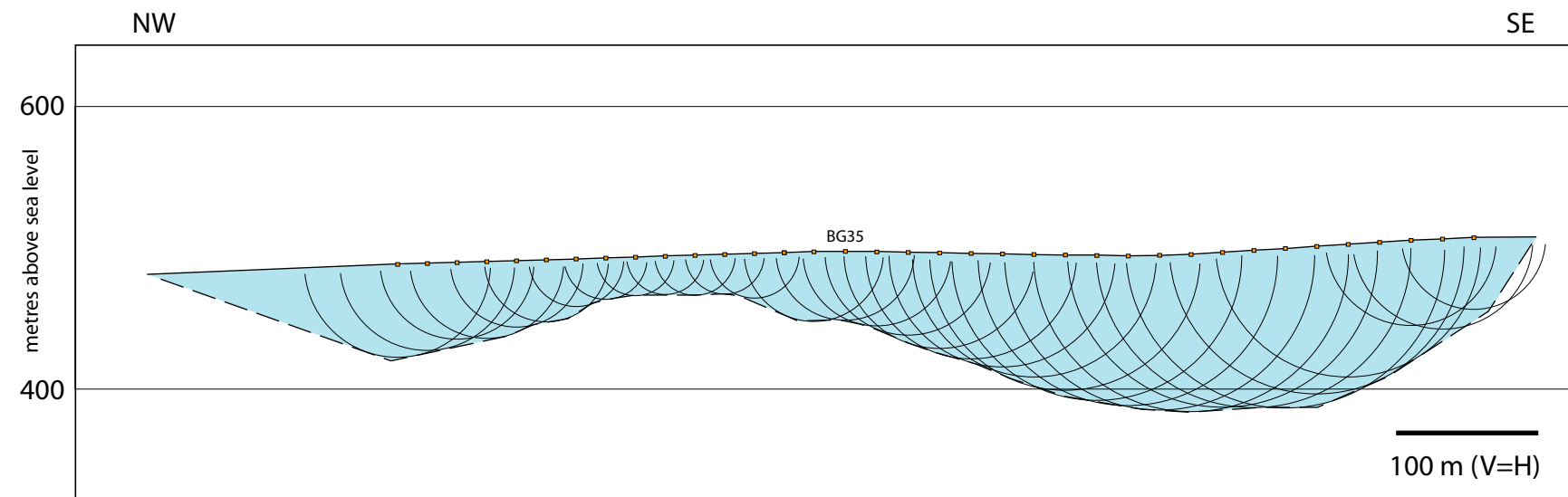


Figure 14. Radio Echo Sounder profile, BG35 (November 2000). Cross section viewed looking down glacier. The figure shows return ellipses for each measurement position, and a “return envelope” representing the glacier bed. The profile position is shown on Fig. 16.

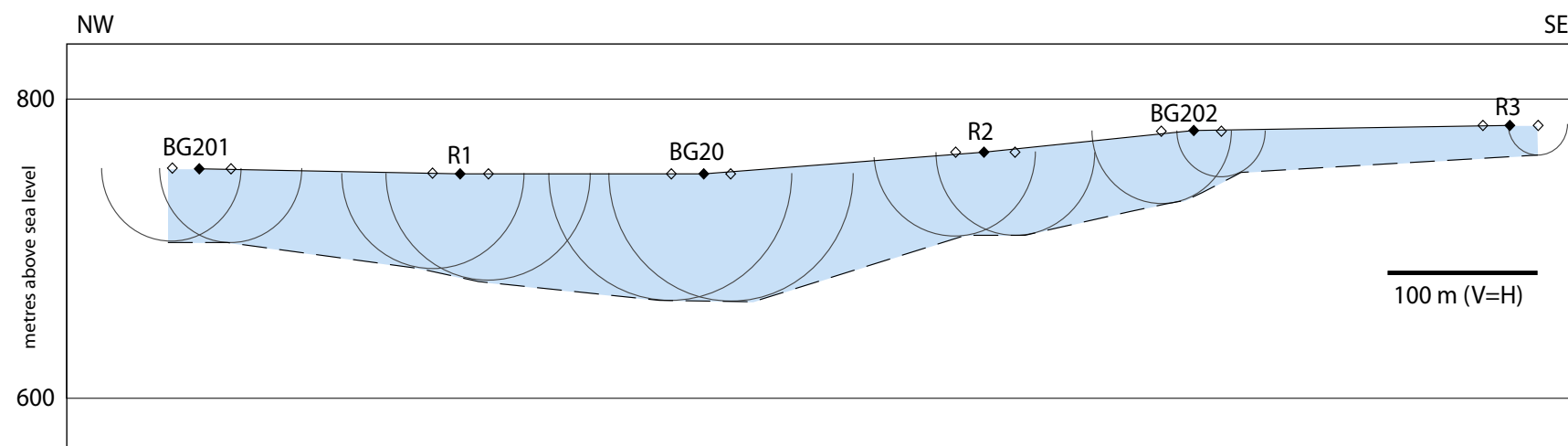
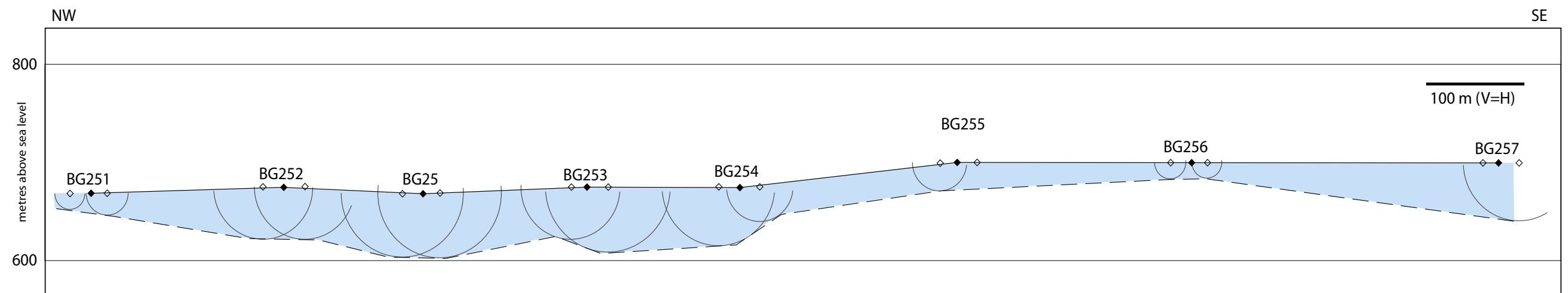


Figure 15. Radio Echo Sounder profiles for BG25 (above) and BG20 (left), 28 January 2004. Cross section viewed looking down glacier. The figure shows return ellipses for each measurement position, and a “return envelope” representing the glacier bed. The profile position is shown on Fig. 16.

8. Digital Elevation Model

The accuracy of mass balance calculations relies to a great extent on the DEM used to determine the area of the glacier. The glacier margin is best defined where it is bounded by rock outcrop. In its accumulation area Brown Glacier blends into an ice field with the adjacent Stephenson and Compton glaciers and the ice divides can only be defined by ice-flow vectors and surface topography.

The glacier boundary used in the 2000 study was defined using a DEM with a 50 m grid derived from *Radarsat* data, which was further refined using kinematic ground DGPS surveys conducted during the 2000 field season (Truffer *et al.*, 2001). The original DEM was adjusted after comparing the accurate heights obtained using DGPS with the *Radarsat* heights.

DGPS kinematic surveys conducted this season were combined with similar surveys from the 2000 season¹, as well as heights acquired using a hand-held *Garmin* GPS V, to create a new DEM for the glacier. The *Garmin*

Table 4. *Surface area calculations for the preferred catchment area (excludes the area of stagnant ice on the north-west margin) using the new DEM, cross sectional areas calculated from RES profiles, average velocity estimates across each profile using measurements made over two field seasons, and the calculated flux and accumulation required to balance that flux through each profile.*

Calculations for preferred catchment area				
		% error	+ error	- error
Total surface area, m ²	2.88E+06	8	3.12E+06	2.64E+06
Area above RES profile				
Above BG35	2.03E+06	9	1.85E+05	1.85E+05
Above BG25	1.23E+06	11	1.39E+05	1.39E+05
Above BG20	8.90E+05	14	1.20E+05	1.20E+05
Cross sectional area, m ²				
BG35 profile	4.70E+04	5	2.35E+03	2.35E+03
BG25 profile	4.09E+04	5	2.04E+03	2.04E+03
BG20 profile	3.88E+04	5	1.94E+03	1.94E+03
Average velocity, m/a				
BG35 profile	27.45	10	30.19	24.70
BG25 profile	21.96	10	24.16	19.77
BG20 profile	19.32	10	21.26	17.39
Flux, m ³ /a water equivalent				
BG35 profile	1.18E+06	16	1.84E+05	1.72E+05
BG25 profile	8.23E+05	16	1.28E+05	1.19E+05
BG20 profile	6.87E+05	16	1.06E+05	9.96E+04
Accumulation required to balance flux, m/a water equivalent				
Above BG35	0.59	15	0.09	0.08
Above BG25	0.67	17	0.11	0.11
Above BG20	0.78	19	0.15	0.14
Balance for areas between profiles, m/a water equivalent				
BG35/BG25	0.45	11	0.05	0.05
BG25/BG20	0.40	10	0.04	0.04

heights were compared with the DGPS heights where survey tracks crossed one another, and were adjusted to match the DGPS survey as closely as possible. Errors of up to 20 m in height were common. This new 40 m grid DEM (Fig. 16) improves significantly on that used in the previous study, providing more accurate definition of ice divides and consequently improved estimates of the glacier area (and incremental areas with elevation). Obvious topographic features poorly represented or absent from the 2000 DEM are much better represented in the 2003 DEM. Appendix A2 shows the distribution of the data used to produce the DEM. Some areas were not traversed, and consequently heights are estimated in these regions.

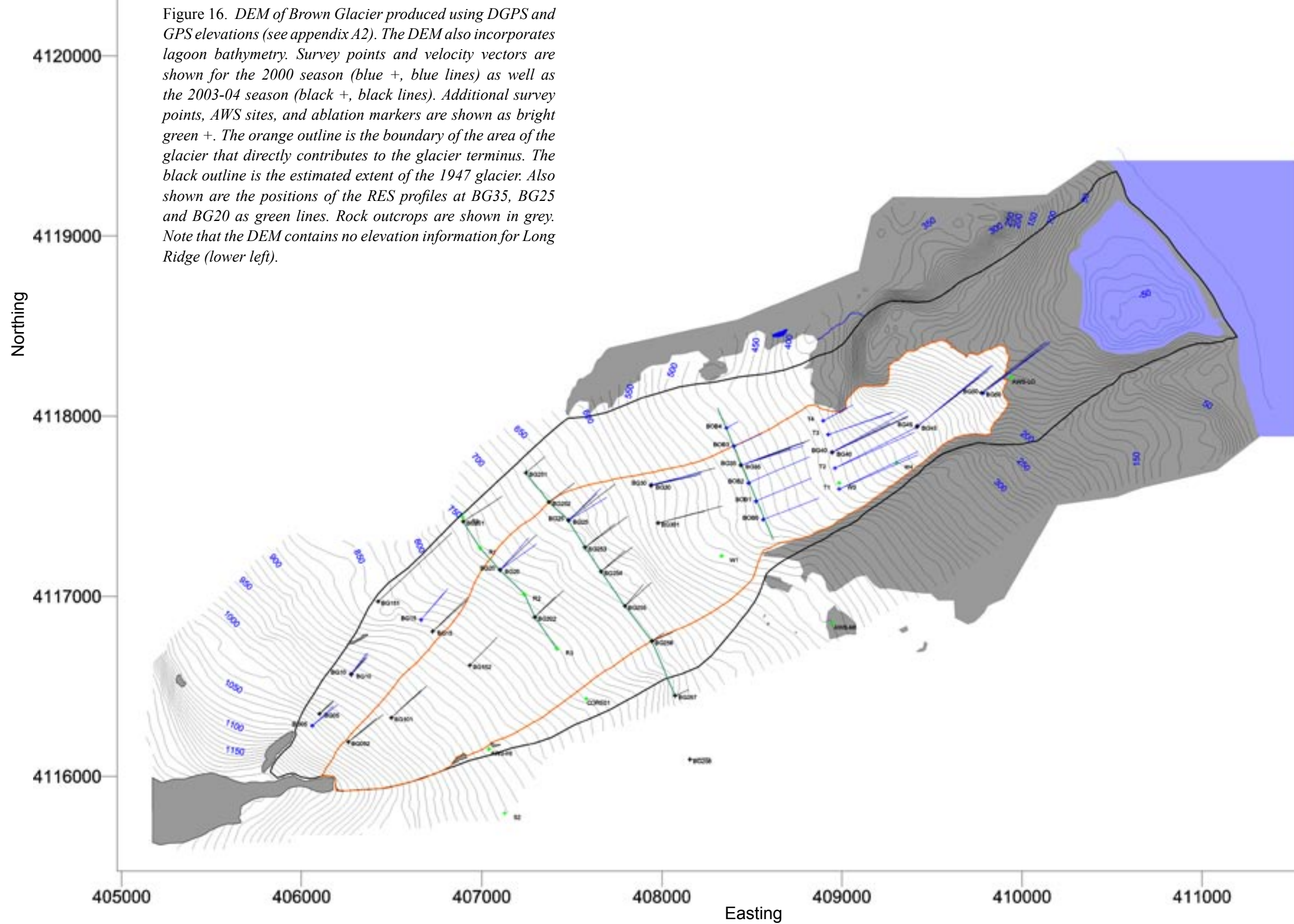
Although the new DEM is a significant improvement on the 2000 DEM, there is still considerable uncertainty associated with defining the actual boundary of the glacier, particularly above 600 m elevation. Even above 400 m elevation, uncertainty is present on the north-western margin of the

Table 5. *Surface area calculations for the maximum catchment area using the new DEM, cross sectional areas calculated from RES profiles, average velocity estimates across each profile using measurements made over two field seasons, and the calculated flux and accumulation required to balance that flux through each profile.*

Calculations for maximum catchment area				
		% error	+ error	- error
Total surface area, m ²	4.42E+06	10	4.45E+05	4.45E+05
Area above RES profile				
Above BG35	3.42E+06	12	4.25E+05	4.25E+05
Above BG25	2.16E+06	15	3.27E+05	3.27E+05
Above BG20	1.65E+06	16	2.71E+05	2.71E+05
Cross sectional area, m ²				
BG35	6.02E+04	4	2.19E+03	2.19E+03
BG25	5.35E+04	8	4.07E+03	4.07E+03
BG20	5.92E+04	6	3.73E+03	3.73E+03
Average velocity, m/a				
BG35	27.45	10	30.19	24.70
BG25	20.11	10	22.12	18.10
BG20	19.31	10	21.24	17.38
Flux, m ³ /a water equivalent				
BG35	1.51E+06	14	2.12E+05	2.01E+05
BG25	9.87E+05	18	1.81E+05	1.66E+05
BG20	1.05E+06	17	1.77E+05	1.64E+05
Accumulation required to balance flux, m/a water equivalent				
Above BG35	0.45	20	0.09	0.08
Above BG25	0.46	18	0.09	0.08
Above BG20	0.65	21	0.14	0.12
Balance for areas between profiles, m/a water equivalent				
BG35/BG25	0.42	23	0.10	0.09
BG25/BG20	-0.12	33	0.04	0.05

¹ Only DGPS surveys on rock areas in 2000 were used in the new DEM, due to the dramatic changes that occurred in the surface elevation of the glacier in the intervening three year period. A few high points from the 2000 survey on ice on the western margin of the glacier were used, where elevation differences of 3-5 metres could be expected.

Brown Glacier, Heard Island



glacier where an area of slow moving or stagnant ice terminates above a high cliff. An ice divide exists between this north-western side of the glacier and the active ice that spills over the cliff and flows to lower levels.

Maximum and minimum areas are defined for the glacier using a combination of measured ice flow vectors and DEM surface down-slope vectors, as well as a bit of “eye-ball guesswork”. The maximum and minimum area boundaries are combined with maximum and minimum cross-sectional areas from the three RES profiles, and ice surface velocity measurements, to determine annual ice flux through each cross-section, and the annual ice accumulation required upslope to balance this flux, with associated errors (Table 4). Most confidence is placed in estimates for the BG35 cross-section.

An additional estimate was made for what is considered to be the entire catchment of Brown Glacier, including the now stagnant north-west margin (Table 5). This also includes a maximum and minimum estimate of the area. As before, uncertainty in defining the glacier margin results from inaccuracies in the DEM (primarily in areas where there is limited survey coverage).

9. Climate

9.1 Automatic Weather Stations

Two *Vaisala* mobile automatic weather stations (MAWS) were deployed at Brown Glacier: one was positioned on bedrock at the glacier terminus (MAWS1: UTM 4118214N, 409941E, 115.5 m a.s.l., Fig. 17a), and the second was sited on a small rock outcrop on the eastern margin of the glacier (MAWS2: UTM 4116151N, 407041E, 920 m a.s.l., Fig. 17c). These units recorded maximum, minimum and average values for temperature, relative humidity, wind speed and direction, and solar radiation, at 10 minute intervals. MAWS2 had a power failure on January 11th, but restarted later, with the correct time of day but incorrect date. This resulted in a gap in the data of 29.16 hours.

Both MAWS1 & 2 were equipped with a tipping bucket rain gauge, however neither worked particularly well: MAWS1 filled with grit, and high winds at MAWS2 resulted in the aluminium arm to which the bucket was attached shearing off at the locating bolt (Fig. 17c). There were a few days of data from each rain gauge, but not overlapping data. The loss of the raingauge mounting arm necessitated the relocation of the T/RH sensor to the pyranometer arm (see Fig. 17a). This configuration was also adopted for MAWS1 at the end of the season.

A third AWS (dubbed “Snowwhite”) equipped with *Aanderaa* instruments was placed adjacent to the 4 m AWS tower left on site in November 2000 (UTM 4116852N, 408948E, 550 m a.s.l., Fig. 17b), and recorded average temperature, relative humidity, and average and maximum wind speed at 10 minute intervals. The wind direction sensor did not work correctly on this unit, and relative humidity was unreliable. This same unit was also deployed on the glacier surface during the 2000 season, and a reassessment of the wind direction data from that period shows that it too is unreliable.

At the end of the field season, MAWS1 was left in place to record weather conditions every three hours. There

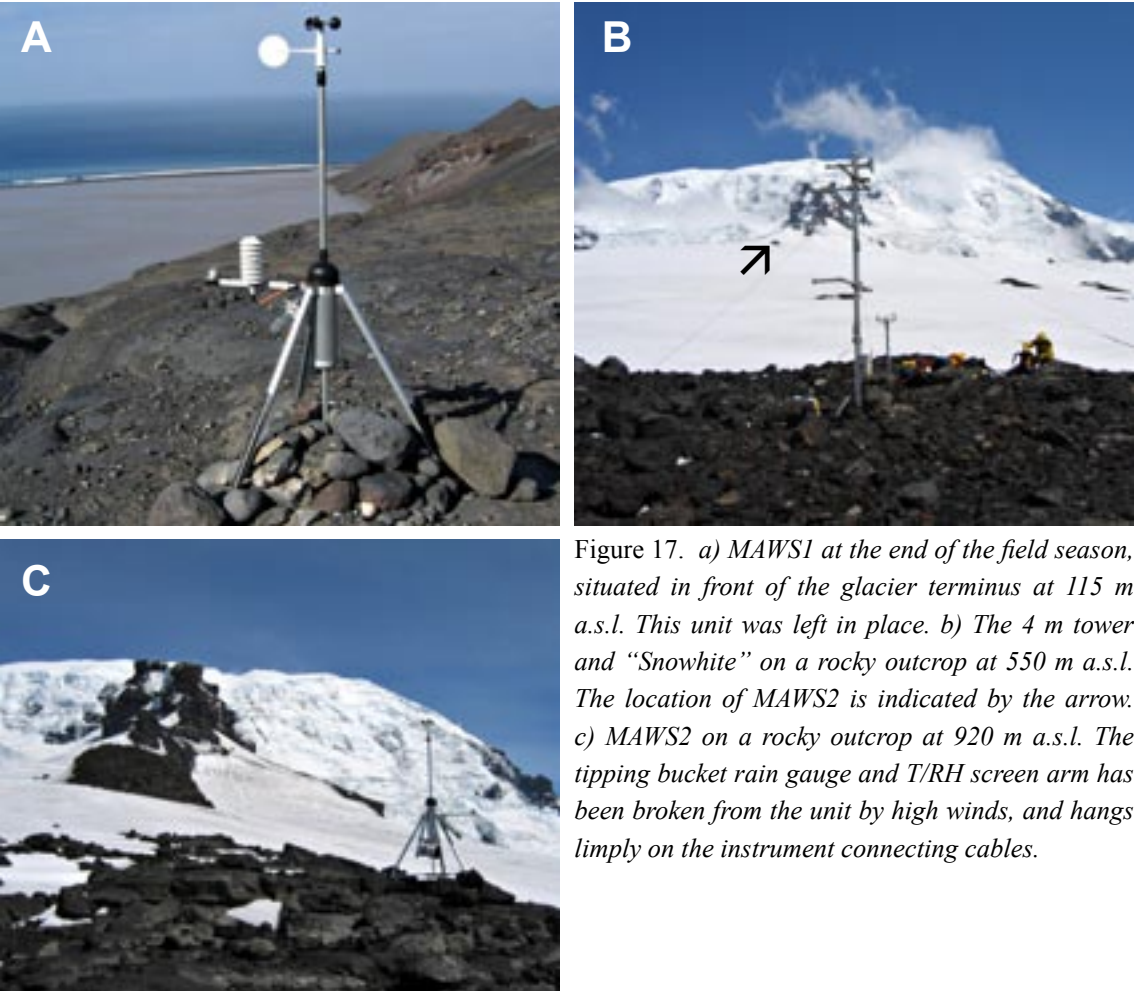


Figure 17. a) MAWS1 at the end of the field season, situated in front of the glacier terminus at 115 m a.s.l. This unit was left in place. b) The 4 m tower and “Snowwhite” on a rocky outcrop at 550 m a.s.l. The location of MAWS2 is indicated by the arrow. c) MAWS2 on a rocky outcrop at 920 m a.s.l. The tipping bucket rain gauge and T/RH screen arm has been broken from the unit by high winds, and hangs limply on the instrument connecting cables.

Table 6. Meteorological data collected during the 2003-04 season. Y = data recorded and reliable; F = data recorded, only intermittently reliable, otherwise missing or suspect; Y* = data recorded throughout the season by BoM data buoys, time between observations variable; y = data logged every half hour by T-TEC datalogger; X = solar radiation data collected constantly at Spit Bay by Johanna Turnbull, Uni of Wollongong. Other data collected by botanists at sites near Scarlet Hill and Aubert de la Rue not listed.

Location	Temp	RH	Wind speed	Wind direct.	Solar rad'n	precipitation	Air Press.
MAWS1, 115m	Y	Y	Y	Y	Y	F	
Snowwhite, 550m	Y	F	Y	F			
MAWS2, 920m	Y	F	Y	Y	Y	F	
Brown Hut						Y	
Spit Bay	Y*				X	Y	Y*
Doppler Hill	y	y					
Jacka Valley	y	y				Y	
Atlas Cove	Y*						Y*
Capsize Beach	y	y				Y	

is a real risk that the unit will lose power during the winter months. It is planned to retrieve this unit at the next available opportunity. The other two units were removed at the end of the field season.

A four metre high aluminium tower with AWS instruments and a battery box was deployed using a helicopter in October 2000 at the 550 m AWS-mid site (Fig. 17b). This unit relayed data using the ARGOS satellite system for 22 months before failing completely in September 2002. Eleven people removed the entire unit on the February 16th 2004.

With field parties deployed around the island, the opportunity existed to make coincident weather observations; primarily on temperature to compare the coincidence or otherwise of Föhn winds, but particularly on precipitation. Detailed records for precipitation only exist for Atlas Cove (1947-54 epoch) and one year from Spit Bay (1993). Each field camp (Jacka Valley, Brown Glacier, Spit Bay, Capsize Beach) recorded daily precipitation, and Jacka Valley and Capsize Beach also deployed a T-TEC temperature – relative humidity logger in a Stephenson screen, logging at 30 minute intervals. The Doppler Hill camp, which was abandoned early in the season, also had a T-TEC logger which was left unshielded and attached to a metal star-picket above a dark sandy surface. Each field party made additional observations of wet and dry bulb temperature, cloud cover, and wind speed, at varying intervals.

9.2 Temperature

Jacka Valley shows the smallest temperature variation, as well as being the coolest of the near sea level field sites (Table 7). Snowwhite and MAWS2 show slight diurnal variation compared with the coastal sites. The greatest variation occurs during protracted Föhn events, when the temperature at the upper sites can even exceed that recorded at sea level. Notable events occurred on the 27 December; 2, 3-6, 8, 9-10, 15, 17-18, 23-24, 29-30 January; and the 2-8 February. Figure 18 shows an example of temperature records from a number of sites during a Föhn event. Of particular interest are the differences in timing for peak temperatures, as well as the pronounced oscillation in temperature between the two upper stations. These oscillations correspond with similar oscillations in wind speed and direction, with the hottest and fastest winds being southerly, and the colder, slower winds being more westerly.

Table 7. Summary of temperature and relative humidity data from T-TEC data loggers deployed at various locations around Heard Island during the 2003-04 summer field season.

	Jacka Valley		Capsize Beach		Doppler Hill	
	T	RH	T	RH	T	RH
Average	4.6	96	6.3	83	6.4	85
Std Dev	1.4	6.6	3.7	12.0	3.1	15.7
Max	10.5	103	20.3	103	18.9	105
Min	0.9	65	0.6	20	0.9	44

Table 8. Temperature summary for January 2004 from the lower to upper glacier sites. Accuracy is ± 0.3 °C at 0 °C. Note the increase in standard deviation with elevation.

Location	MAWS1	Snowwhite	MAWS2
Average temperature, January	4.96	1.72	0.51
Max temperature	15.02	15.53	12.92
Min temperature	0.28	-3.43	-5.50
Standard deviation	2.61	3.45	4.36
Relative humidity	77	-	-

Table 9. Wind Speed summary for January 2004 from the lower to upper glacier sites. Sensors for MAWS1 & 2 were Vaisala combined wind sensors WMS302 and WMS302. Snowwhite used Aanderaa wind speed sensor 2740. Anemometer accuracy ± 0.3 m.s⁻¹ for wind speeds ≤ 10 m.s⁻¹, <2% for wind speeds > 10 m.s⁻¹.

Location	MAWS1	Snowwhite	MAWS2
Average monthly wind speed (m.s ⁻¹)	4.5	10.1	10.6
Max 10 min avg wind speed (m.s ⁻¹)	10.4	33.1	36.8
Max gust (m.s ⁻¹)	32.5	52.3	62.5
Dominant direction	NW, WNW	-	W, NW

Table 10. Correlation coefficients for temperature, wind speed, and temperature and wind speed between sites. These correlation coefficients use daily averages over the observation period. The highest correlation for temperature and wind speed considered independently occurs (predictably) between the stations closest to each other (ie the lowest and intermediate; the intermediate and highest). Note that the best correlation between wind speed and temperature is obtained between the MAWS1 temperature and MAWS2 wind speed (ie a high temperature day at MAWS1 correlates with a high wind speed day at MAWS2: mainly due to the influence of Föhn events).

Correlation coefficients between sites, daily averages

MAWS1 - 2	MAWS1 - Snowwhite	MAWS2 - Snowwhite
temperature correlation		
0.811	0.947	0.924
wind speed correlation		
0.864	0.898	0.946
wind speed – temperature correlation (lower site T, upper site wind)		
0.585	0.446	0.481

9.3 Wind Speed

There is a large variation in wind speed with elevation (Fig. 19), most noticeably between MAWS1 and Snowwhite (Table 9). The average monthly wind speed at 550 m was more than double that recorded at 115 m in January. This is mainly due to the protection afforded to the lower glacier by the bounding cliffs. The maximum recorded gust at MAWS2 of 225 km.h^{-1} exceeds all earlier records, which is not surprising, considering the elevation and exposure of the site. Dominant directions are westerly (18% of the time), north-westerly (11%), and south-westerly (7%) at MAWS2, and mainly west-northwesterly to north-northwesterly at MAWS1 (35%).

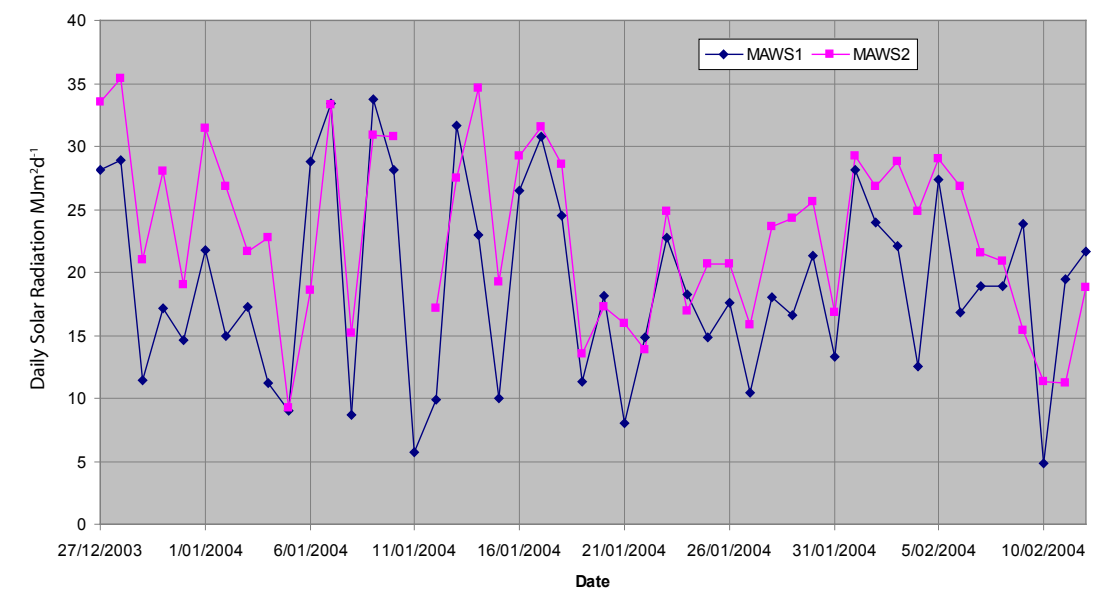
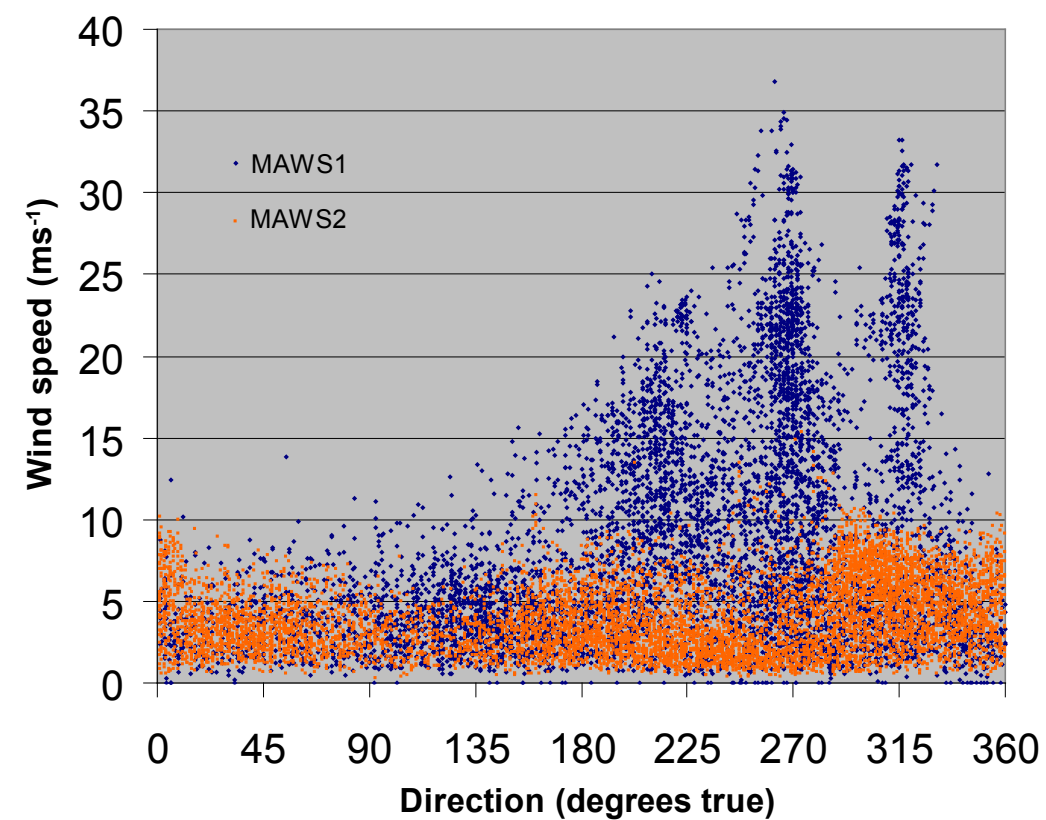
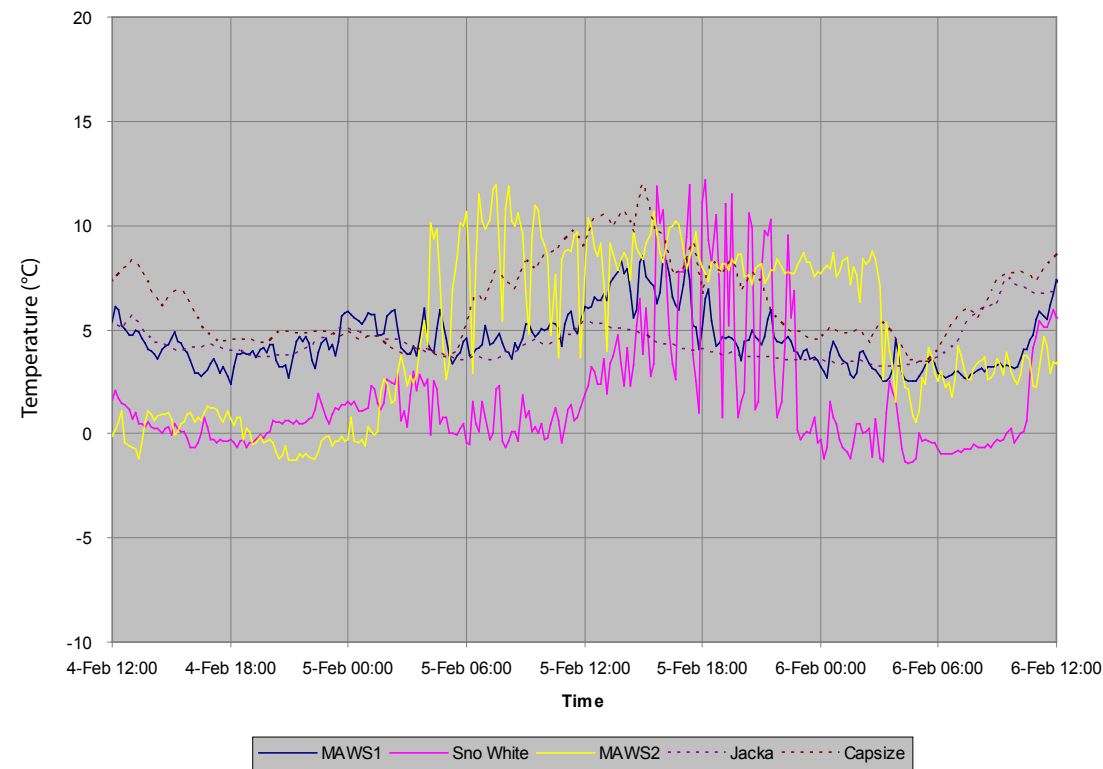
9.4 Solar Radiation

Daily solar irradiance was measured at MAWS1 and MAWS2 using a *Kipp & Zonen* CM3 pyranometer, with a spectral sensitivity from 0.3 to 3 microns. Readings are 10 minute averages, with maximum and minimum values also recorded. Solar radiation was greater at MAWS2 than MAWS1 on 37 out of 47 days (Fig. 20). MAWS1 received 86% of the solar radiation that MAWS2 received. This difference is mainly attributed to low level cloud between the stations.

Figure 18 (upper left). A detailed snapshot of the temperature data either side of the 5th of February when a particularly long-lived Föhn wind was present. Note the extreme temperature recorded by MAWS2, as well as the oscillation between warm and cool periods at both the upper stations.

Figure 19 (lower left). A plot of 10 minute average wind direction and speed, for MAWS1 and MAWS2. Note the dramatic difference in wind speed between the two sites, and the trimodal distribution of wind direction at the upper site.

Fig. 20 (below). Daily solar radiation at MAWS1 and MAWS2.



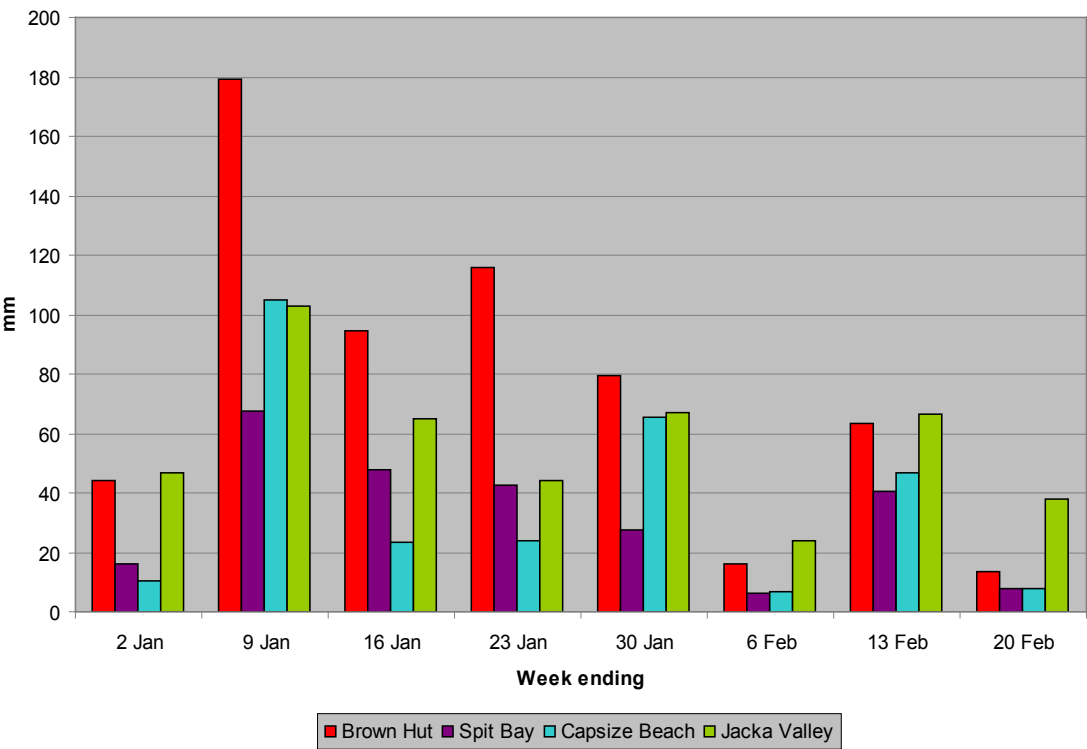
9.5 Precipitation

All sites recorded very high rainfall totals in comparison to earlier records. The only locality that was close to being re-occupied, and is directly comparable with earlier records, was at Spit Bay (although the exact location of the rain gauge in 1992-93 was not known). Not only is temperature highly variable with location, so is rainfall. The most obvious illustration of this is a comparison of the records from Spit Bay and Brown Hut: within sight of each other and separated by only 5 km, with a 250% difference in rainfall.

Table 11. *Summary of precipitation records for January, various years and locations, Heard Island. Amount in mm.*

AVERAGE for Jan recorded at Atlas Cove 1948-54	139.3
MAX for Jan recorded at Atlas Cove 1948-54	168.9
MIN for Jan recorded at Atlas Cove 1948-54	102.6
Jacka Valley Jan 2004	298.9
SPIT BAY Jan 1993	128.0
Spit Bay Jan 2004	189.9
Brown Hut Jan 2004	482.9
Capsize Beach Jan 2004	221.5

Figure 21. *Total weekly precipitation for 2003-04 field sites, Heard Island*

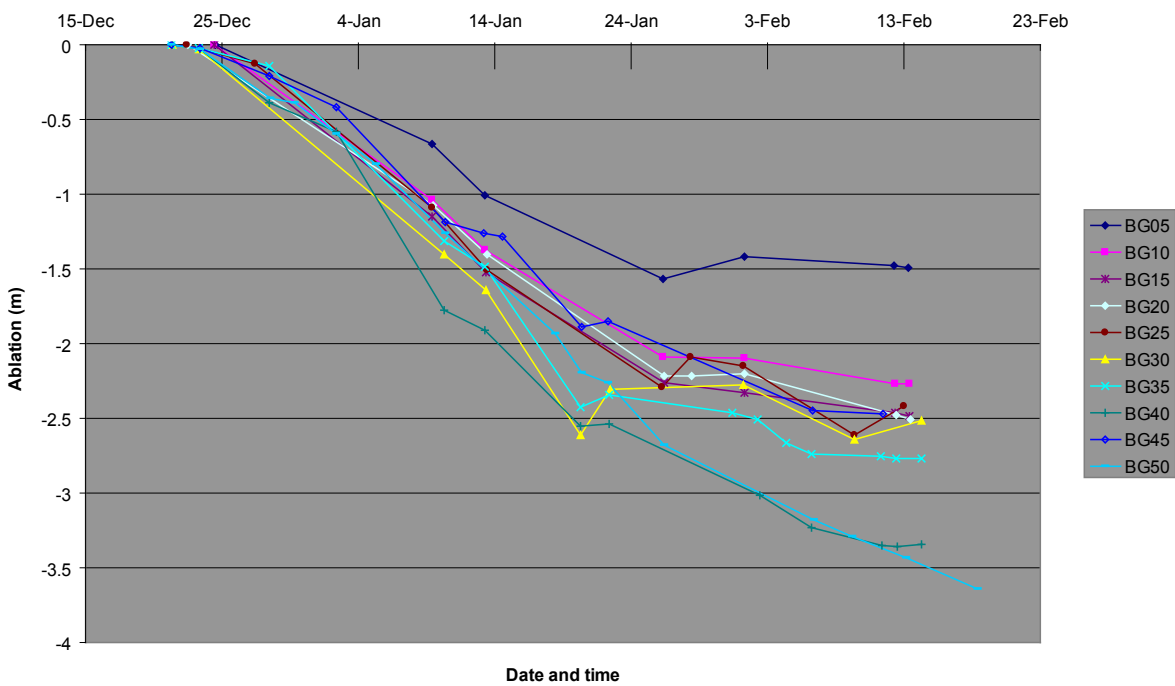


It was hoped that the MAWS equipped with tipping bucket rain gauges would be able to give us our first real indication of variation in precipitation with elevation. The deployment this season was less than successful, however, with both units failing due to high winds and blowing grit. Some data were collected over a number of days. Estimates for precipitation increase with elevation using MAWS2 compared with Brown Hut range between 0.06 and 0.19 mm.m⁻¹.

10. Ablation measurements

Each fibreglass survey pole was visited several times, and pole height above the snow/ice surface measured (Fig. 22). In addition, sonic rangiers were erected on an aluminium bi-pole – cross-brace arrangement at BG50 and BG35 (Fig. 23a-c), measuring the distance to surface every half hour. The BG50 sonic ranger recorded faultlessly, although ablation values derived from pole measurements and the sonic ranger diverge early in the season. The site of the pole and sonic ranger were separated by about 30 m, and as the season progressed the surface became extremely rough and crevassed (Fig. 23d, e). The ablation rate between the two sites may have been influenced by the proximity to crevasses and the dwindling size of some seracs, as well as variations in albedo. Operation of the sonic ranger at BG35 became intermittent, and pole measurements provided a better record of ablation at this site late in the season.

Figure 22. *Ablation (with minor accumulation) records for the field season from the main survey sites, Brown Glacier. This record takes no account of the difference in surface density, explaining why many of the higher sites (with a snow/firn surface) record as much or more ablation than lower sites (on ice).*





11. Ice coring and crevasse sampling

Ice, snow, and water samples were collected from the Brown and Stephenson Glaciers for $\delta^{18}\text{O}$, and ion chemistry analysis. Samples were collected from three crevasse sites and two hand-cored drill sites (12.1 l), 20 surface samples of freshly fallen snow (1.6 l), rain water (0.12 l) and a sample from Brown Lagoon (0.54 l). Hand coring (Fig. 24a-c) was abandoned in favour of crevasse wall sampling (Fig. 24d-e), due to the difficulty of using the ice corer in the wet snow pack. An ice screw used for sampling was lost in the wall of the crevasse near BG152.

Samples were analysed at the Antarctic Climate & Ecosystems (ACE) CRC labs at the University of Tasmania (S. Donoghue, PhD study) for $\delta^{18}\text{O}$ and ion chemistry. The remaining water is stored in the freezers in the ice mechanic lab at the ACE CRC.

It was hoped that the $\delta^{18}\text{O}$ values would reveal a seasonal cycle of accumulation. If the cycles are present, they are obscured by smoothing due to water percolation and fractionation. The samples were also analysed for Cl, SO_4 , NO_3 , Mg, Ca, and Na, which may provide a better estimate of the average annual accumulation on Brown Glacier (data is presented on the data CD: refer to Appendix A1).

12. Lagoon surveys

Brown, Compton, Stephenson and Winston Lagoons were surveyed from an 18 foot *Zodiac* IRB, with depths determined using a *Garmin* “Fish Finder” depth sounder. The transponder was attached to a length of PVC pipe, held vertically and braced against the side of the *Zodiac*. Position was determined using a hand held *Garmin* GPS V. Positional accuracy is within 10 m. At a number of locations, the indicated depth was checked against a weight lowered on a 30 m tape measure, and found to be within 1 m. Depths greater than 30 m could not be checked. Considering possible variations in water temperature, suspended sediment in the water column, and inconsistencies in levelling the transponder, depth soundings are considered to be accurate to $\pm 5\%$ at depths greater than 30 m.

Contoured bathymetric charts are presented for the lagoons in Appendix A3.

Figure 23. a) BG35 with various instruments deployed, on the 10th January 2004. In the foreground is the albedometer, with the grey Pelican case containing the logging electronics. This unit failed to log despite repeated attempts to reset it. Behind, is the sonic ranger; and to the left can be seen the fibreglass survey marker. b) BG35 on the 20th January. The albedometer has completely melted out and the sonic ranger is nearly 3 m above the surface. c) BG50 on the 5th January: a very wet day. To the left of the sonic ranger can be seen the Trimble GPS antenna mounted on an aluminium pole, with the GPS gear housed in the yellow case to the right. This unit recorded position up to four times a day towards the end of the season. Photo M. Truffer. d) BG50 survey marker location which became increasingly crevassed and difficult to access as the season progressed. 22nd January. e) The view from the east terminus towards BG50, with the sonic ranger visible on the ice horizon. Access was along a few narrow ice-ribs. Note the low albedo of the lower ice. 9th February.

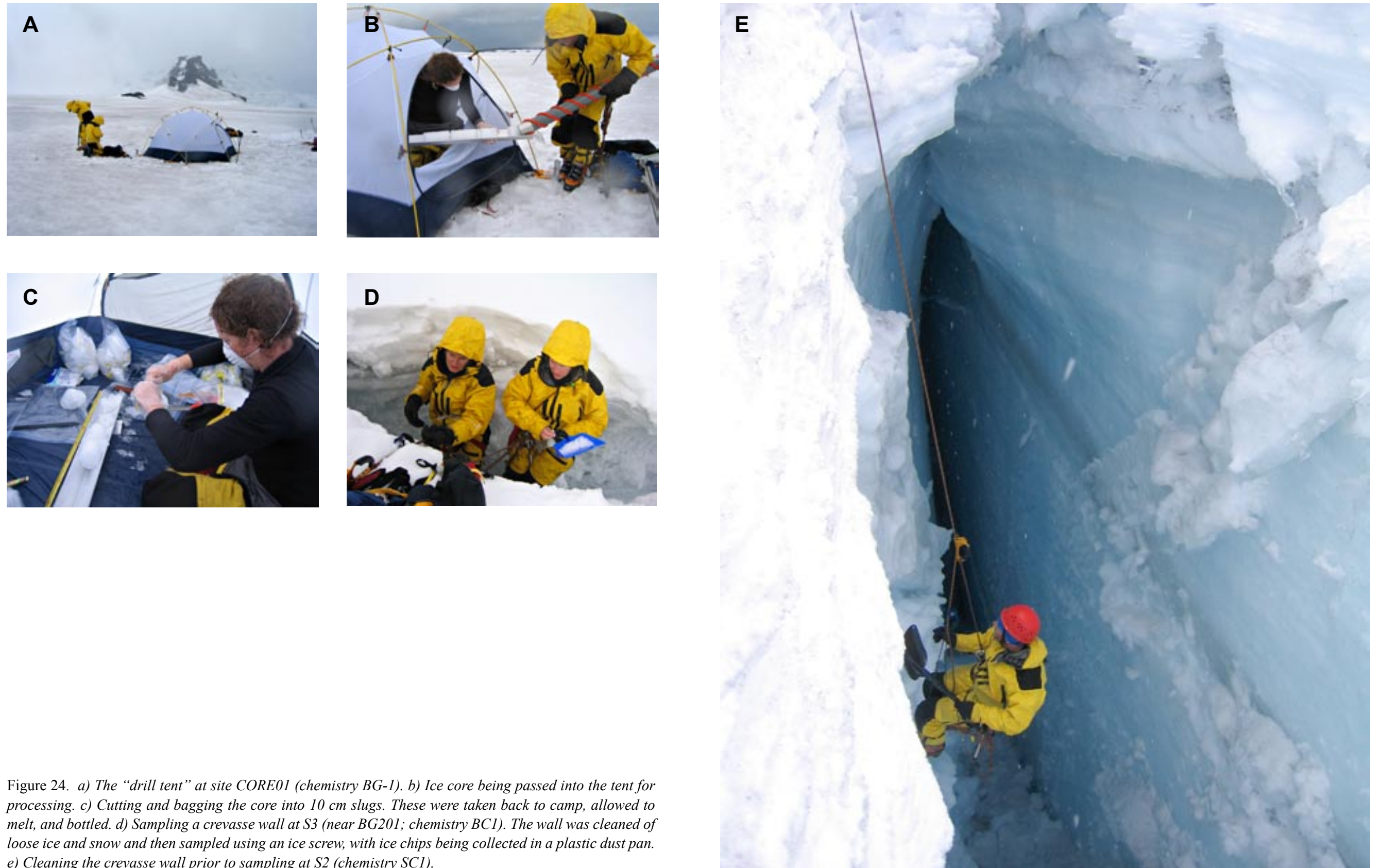


Figure 24. a) The "drill tent" at site CORE01 (chemistry BG-1). b) Ice core being passed into the tent for processing. c) Cutting and bagging the core into 10 cm slugs. These were taken back to camp, allowed to melt, and bottled. d) Sampling a crevasse wall at S3 (near BG201; chemistry BC1). The wall was cleaned of loose ice and snow and then sampled using an ice screw, with ice chips being collected in a plastic dust pan. e) Cleaning the crevasse wall prior to sampling at S2 (chemistry SC1).

13. Catchment Runoff

The amount of water entering Brown Lagoon was calculated on 31 December. Cross-sectional profiles of the creeks entering the lagoon were measured and flow velocities estimated using a flow meter, to calculate flow rate in $\text{m}^3\text{sec}^{-1}$. This was compared with the flux calculation for water flowing out of the lagoon mouth, the intention being to only monitor the lagoon mouth as an indicator of total catchment outflow. The two flow rate values differed by 9%, with outflow ($1.475 \text{ m}^3\text{sec}^{-1}$) being higher than inflow ($1.353 \text{ m}^3\text{sec}^{-1}$).

14. Accumulation and Ablation indicators

Eight magnets were left in the accumulation zone on the 13th February, approximately 30-60 cm below the surface (denoted by (m) in Fig. 5, and Table 1). The magnets are 2.54 cm diameter and 25.4 cm long, encased in a 28 cm long PVC pipe. Our intention is to locate the magnets using a magnetometer during the next field season to determine average ice flow velocity and accumulation rate over the intervening period. The magnets will probably be buried at depths of up to 6 m, and will not be recoverable.

Five ablation wire markers (Fig. 25; W1-5, Fig. 5, and Table 1) were emplaced in Brown Glacier using a portable steam drill, at a depth of approximately 10 m. Wire W1 was emplaced on the 4th February, and the others on the 6th February. Each hole is approximately 3 cm in diameter. A 610 g lead fishing weight is attached to a 15 m length of 12 gauge single-cored copper electrical wire sheathed in PVC (coloured blue). This was lowered down the hole, with 1 m of excess wire left above the surface. The hole is marked for relocation using six 40 cm steel welding rods arranged in a tetrahedron, which sits on the surface of the glacier and is attached to the excess wire from the hole. Repeat measurements of the length of wire exposed (less 1 m) measures the amount of ice melting at each location (and thereby the ablation rate). Over a period of years the wire, and ultimately the weights themselves, will return to the surface as the ice melts and flows down slope, allowing their recovery and eventual removal. The welding-rod tetrahedron remains over the hole, moving with the surface.

Unlike the magnetic markers, these wire markers are recoverable in the short term. The coil of wire that accumulates at the surface will not blow away or pose a threat to wildlife, as the weight keeps it in place.

15. General Observations

A comparison between that portion of the high resolution *Quickbird* image covering Stephenson Lagoon (taken on January 12th 2003) and the lagoon survey track revealed that the front of Stephenson Glacier had retreated ~200 m in one year. As had been predicted three years previously by many people working on the island, Stephenson Lagoon became continuous to the south coast, and navigable by IRB (see Appendix A3).

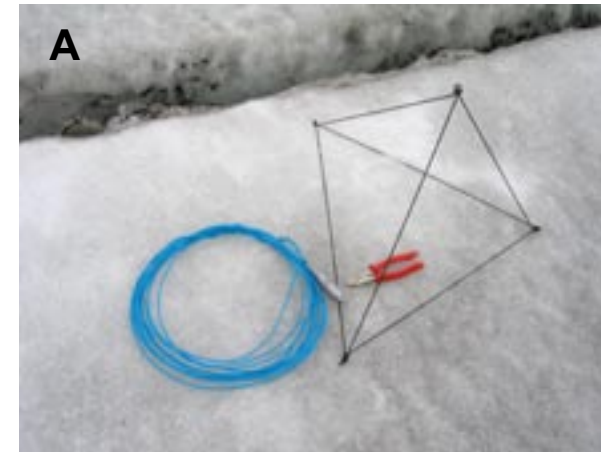


Figure 25. a) Components of the ablation wire markers: a tetrahedron made from steel welding rod, a 610 g lead weight, and a length of PVC-sheathed copper wire. Pliers for scale. b) Using a portable steam drill to make a 3 cm diameter, ~10 m deep hole at site W1.

References

- Allison, I.F. 1980. A preliminary investigation of the physical characteristics of the Vahsel Glacier, Heard Island. *ANARE Scientific Reports Series A* (4); Glaciology. Publication 128. Canberra, Australian Government Publishing Service. 31 pp.
- Allison, I. & Keage, P.L. 1986. Recent changes in the glaciers of Heard Island. *Polar Record*, 23 (144), 255-271.
- Arendt, A.A., Echelmeyer K.A., Harrison W.D., Lingle C.S., & Valentine, B. 2002. Rapid wastage of Alaska glaciers and their contribution to rising sea level. *Science*, 297, 382-386.
- Budd, G.M. 1964. The ANARE 1963 expedition to Heard Island. *ANARE Reports*, Series A (1); Narrative. Publication 74. Melbourne, Commonwealth of Australia. 53 pp.
- Budd, G.M. 2000. Changes in Heard Island glaciers, king penguins and fur seals since 1947. *Papers and Proceedings of the Royal Society of Tasmania*, 133 (2), 47-60.
- Budd, W.F. & Allison, I.F. 1975. An empirical scheme for estimating the dynamics of unmeasured glaciers. *Snow and Ice Symposium. (Proceedings of the Moscow Symposium, August 1971)*: I.A.S.H. Publication 104, 246-256.
- Jacka, T.H. & Budd, W.F. 1998 Detection of temperature and sea-ice extent changes in the Antarctic and Southern Ocean, 1949 - 1996. *Annals of Glaciology* 27, 553-559.
- Kirkwood, R. & Burton, H. 1988. A bathymetric survey of Winston Lagoon. *ANARE News*, 54, 8.
- Lambeth, A.J. 1951. Heard Island - Geography and glaciology. *Journal and Proceedings of the Royal Society of New South Wales for 1950*, 84 (2), 92-98.
- Narod, B.B. & Clarke, G.K.C 1994. Miniature high-power impulse transmitter for radio-echo sounding. *Journal of Glaciology*, 40 (134), 190-194.
- Quilty, P.G. & Wheller, G.E. 2000. Heard Island and the McDonald Islands: a window into the Kerguelen Plateau. *Papers and Proceedings of the Royal Society of Tasmania*, 133 (2), 1-12.
- Ruddell, A. & Allison, I. 1998. The sensitivity of glaciers at Heard Island to climate change, and their recent response. In Selkirk, P. (ed.): Heard Island wilderness reserve: Reports on natural science and cultural heritage research, *ANARE Research Notes* 101, Australian Antarctic Division, Kingston, Tasmania, 32.
- Ruddell, A.R. In press. An inventory of present glaciers on Heard Island and their historical variation. In: Green, K. (ed.) Heard Island: a biogeographical isolate. Surrey Beatty and Sons, Chipping Norton.
- Taljaard, J. J. & van Loon, H. 1984. Climate of the Indian Ocean south of 35°S. In: H. van Loon (ed.) *Climates of the Oceans. World Survey of Climatology*, 15, 505-601. Elsevier, Amsterdam.
- Truffer, M., Thost, D. & Ruddell, A. 2001. The Brown Glacier, Heard Island: its morphology, dynamics, mass balance and climatic setting. *Antarctic CRC Research Report*, 24. Cooperative Research Centre for the Antarctic and southern Ocean Environment, University of Tasmania, Hobart. 27 pp.
- Williams, G.D. 1998. The response of Heard Island glaciers to climate change. Unpublished honours thesis, Institute for Antarctic and Southern Ocean Studies, University of Tasmania, Hobart. 113 pp.

Appendices

A1 Description of data files available on CD

Ablation folder

survey_canes_ablation.xls

Excel file with the measured height of each survey wand above snow/ice surface for the field season.

BG35_pinger.xls

Excel file with sonic ranger data for BG35.

BG50_pinger.xls

Excel file with sonic ranger data for BG50.

Chemistry folder

Ion_Chemistry.xls

Excel file with analyses of chloride, sulphate, nitrate, Mg, Ca, Na of samples collected from crevasses and cores.

Oxygen_isotopes.xls

Excel file with $\delta^{18}\text{O}$ analyses of samples collected from crevasses and cores.

DEM folder

dem_2003.grd

ASCII file with the 2003 DEM grid, generated using Golden Software *Surfer* v7.0. Header file format is:

id	The identification string DSAA that identifies the file as an ASCII grid file.
nx ny	nx is the integer number of grid lines along the X axis (columns) ny is the integer number of grid lines along the Y axis (rows)
xlo xhi	xlo is the minimum X value of the grid xhi is the maximum X value of the grid
ylo yhi	ylo is the minimum Y value of the grid yhi is the maximum Y value of the grid
zlo zhi	zlo is the minimum Z value of the grid zhi is the maximum Z value of the grid
grid row 1	
grid row 2	
grid row 3...	these are the rows of Z values of the grid, organized in row order. Each row has a constant Y coordinate. Grid row 1 corresponds to ylo and the last grid row corresponds to yhi. Within each row, the Z values are arranged from xlo to xhi. Blanked grid nodes are given a Z value of 1.070141E+038. Rows are 39.855 m apart, Columns are 40 m apart.

dem_11.xls

Excel file with all points used to calculate the dem_2003.xls grid (refer to A2).

The folder also contains high resolution jpeg images of Fig. 16 and the data distribution figure (A2).

Lagoon bathymetry folder

Folder containing Excel files with Easting, Northing (acquired using *Garmin* GPS V; WGS84, UTM zone 43) and depth (acquired using *Garmin* “Fishfinder” depth sounder) for each lagoon surveyed. Also high resolution jpeg images of bathymetric maps reproduced in appendix A3.

Meltwater folder

Contains an excel file with stream profiles and flux calculations, and repeat measurements of Brown Lagoon outflow stream. Also contains jpeg photos of three of the inflow streams, and an image showing their location using the *Quickbird* satellite image for reference.

Photos folder

Contains jpeg digital photos used in this report.

Report folder

HI_data_report_screen.pdf

HI_data_report_print.pdf

This data report is reproduced as both a low and high resolution Adobe *Acrobat* PDF file, for on-screen viewing and printing respectively.

RES folder

BG_35_2000.xls

Excel file with RES data for the BG35 profile, 2000 field season.

RES.xls

Excel file with RES data for the BG25 and BG20 profiles, 2003-04 field season.

Surveys folder

all_survey_points.xls

Excel file with the position of the survey markers and additional points.

daily_position_BG50.xls

Excel file with daily (occasionally more frequent) DGPS position near BG50

kinematic_2000.xls

Excel file with all DGPS kinematic surveys conducted during the 2000 field season.

kinematic_surveys.xls

Excel file with all DGPS kinematic surveys conducted during the 2003-04 field season.

surface_site_surveys.xls

Excel file with the DGPS repeat survey positions of each survey site, for the 2000 and 2003-04 field seasons, and velocity calculations for each epoch.

terminus_surveys.xls

Excel file with the DGPS surveys of the position of the glacier terminus.

Weather observations folder

AANDERAA_data.xls

Excel file with data recorded by the automatic weather station at 550 m a.s.l.

all_data_comparison.xls

Excel file with compilation and graphs of all data from each of the Brown Glacier AWS.

MAWS1_data.xls

Excel file with data recorded by the automatic weather station at 115 m a.s.l.

MAWS2_data.xls

Excel file with data recorded by the automatic weather station at 920 m a.s.l.

precipitation_record.xls

Excel file with rain gauge records from Jacka Valley, Brown Hut, Spit Bay and Capsize Beach.

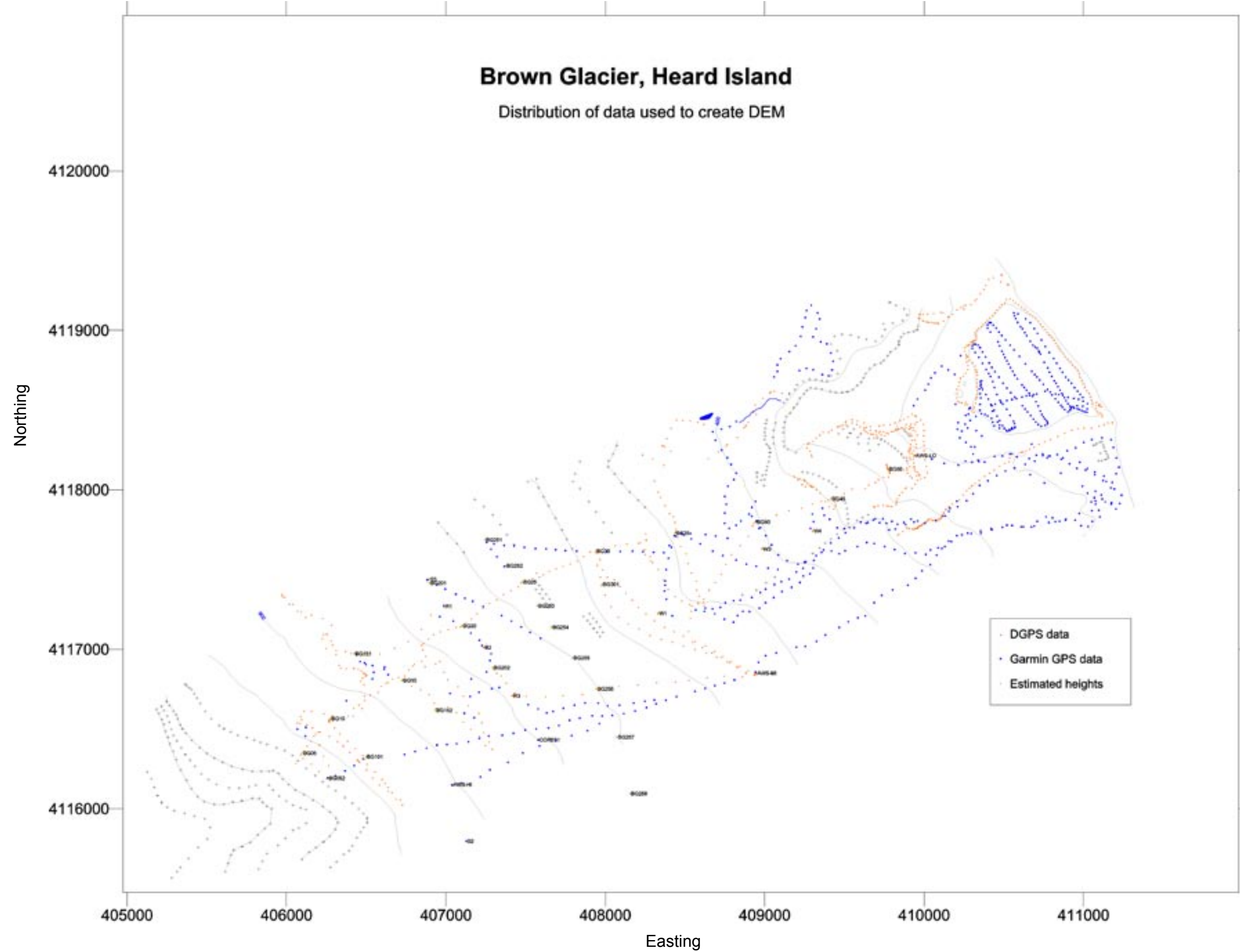
ttec_T_RH_data.xls

Excel file containing temperature and relative humidity data from the three T-TEC loggers, deployed at Jacka Valley, Capsize Beach, and Doppler Hill.

wx_station_photos_folder

Folder containing jpeg photos of each of the weather stations, as well as the field camp observing sites. Missing is Spit Bay.

A2 Digital Elevation Model



A3 Lagoon Bathymetry

Figure A3.1 Bathymetry of Brown Lagoon.

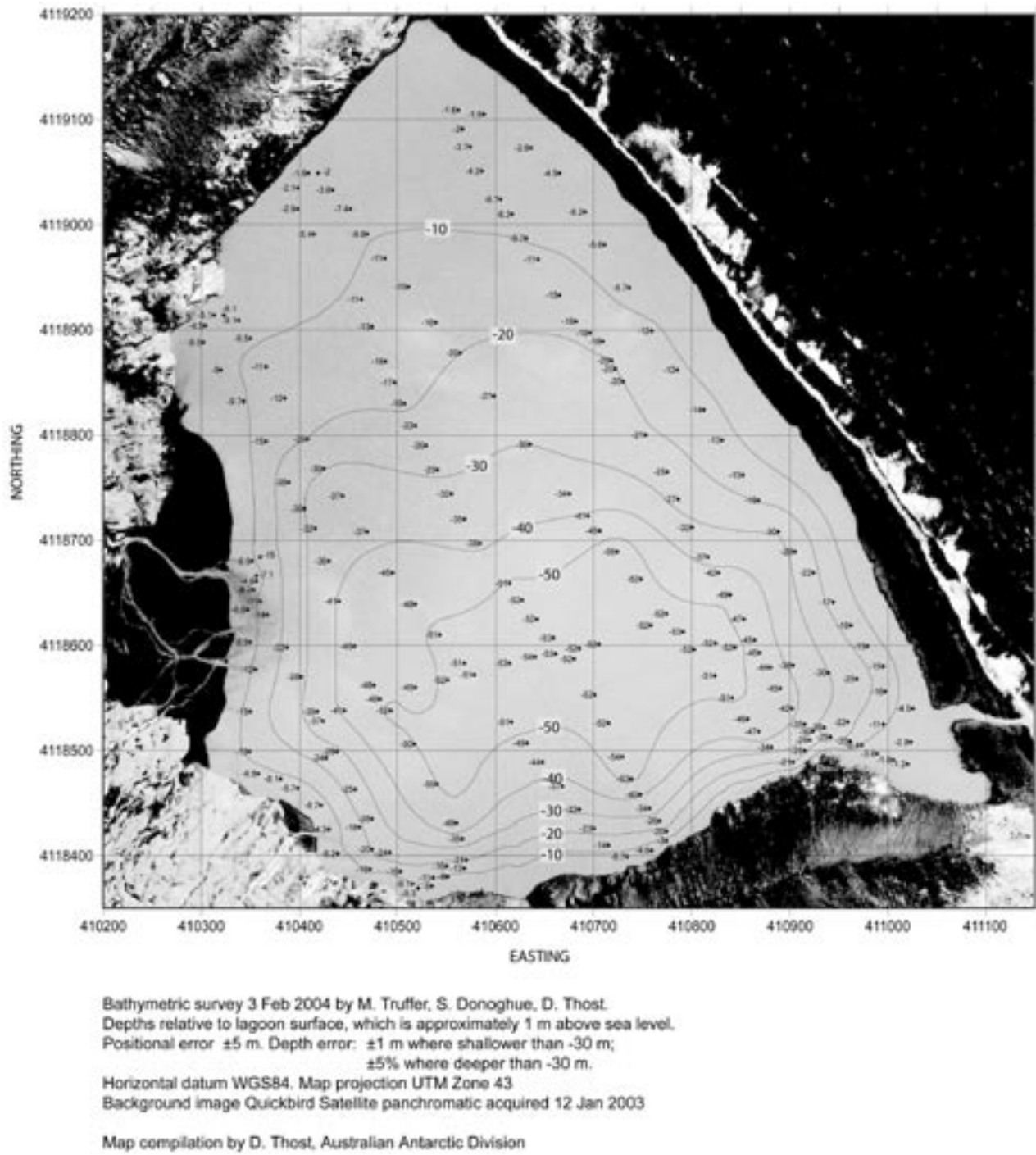
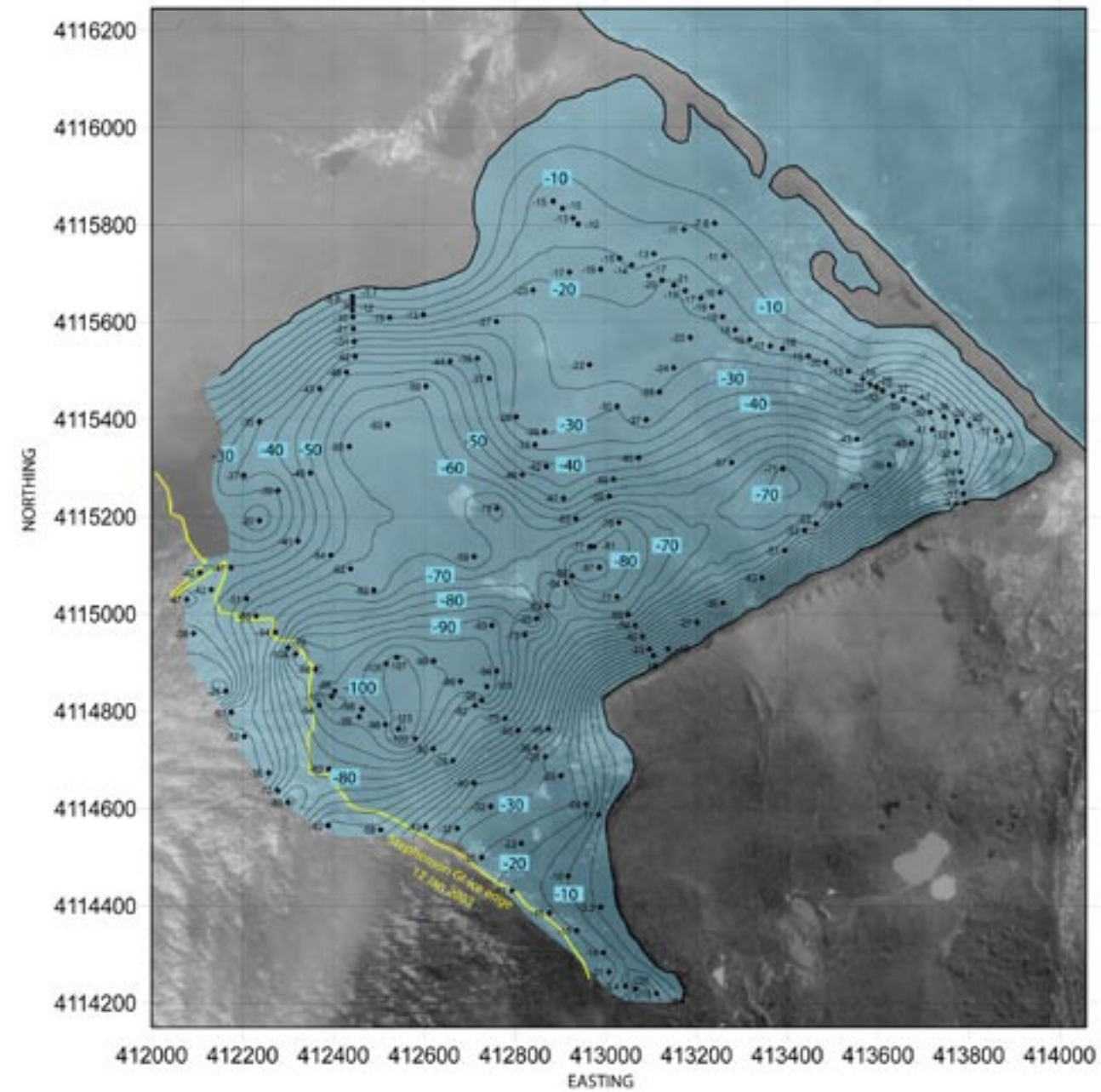


Figure A3.2 a) Brown Lagoon viewed looking northeast from ~200 m a.s.l. on Brown Glacier.

b) During periods of high rainfall the streams feeding the lagoon can become impassable and the delta becomes a floodplain. January 5th, 2004.



Figure A3.3 Bathymetry of Stephenson Lagoon.



Bathymetric survey 17 Jan and 16 Feb 2004 by R. Clifton, S. Donoghue, D. Thost.
 Depths relative to lagoon surface, which is approximately at sea level.
 Positional error: ± 5 m. Depth error: ± 1 m where shallower than -30 m;
 $\pm 5\%$ where deeper than -30 m.
 Horizontal datum WGS84. Map projection UTM Zone 43
 Background image Quickbird Satellite panchromatic acquired 12 Jan 2003
 Map compilation by D. Thost, Australian Antarctic Division

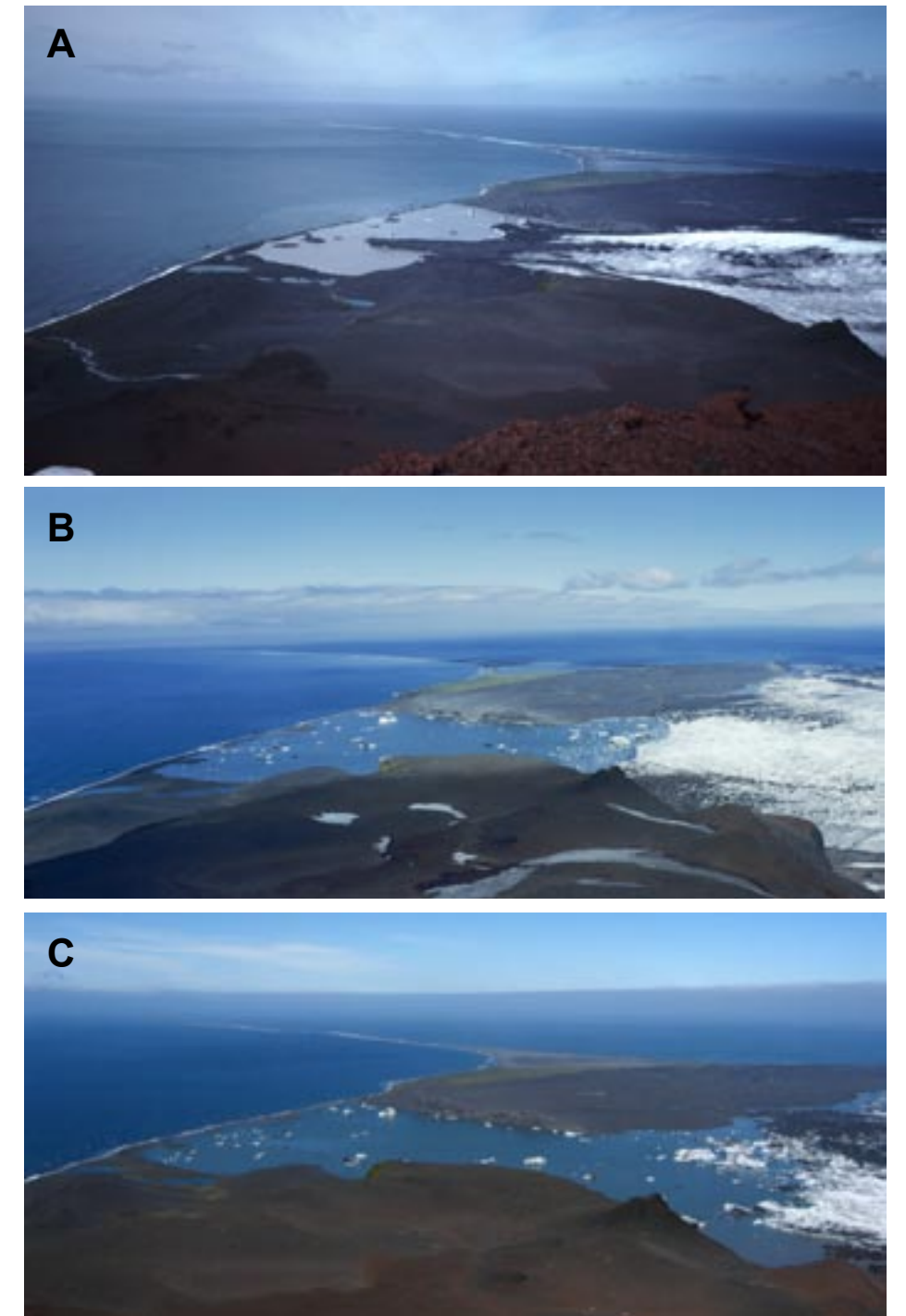
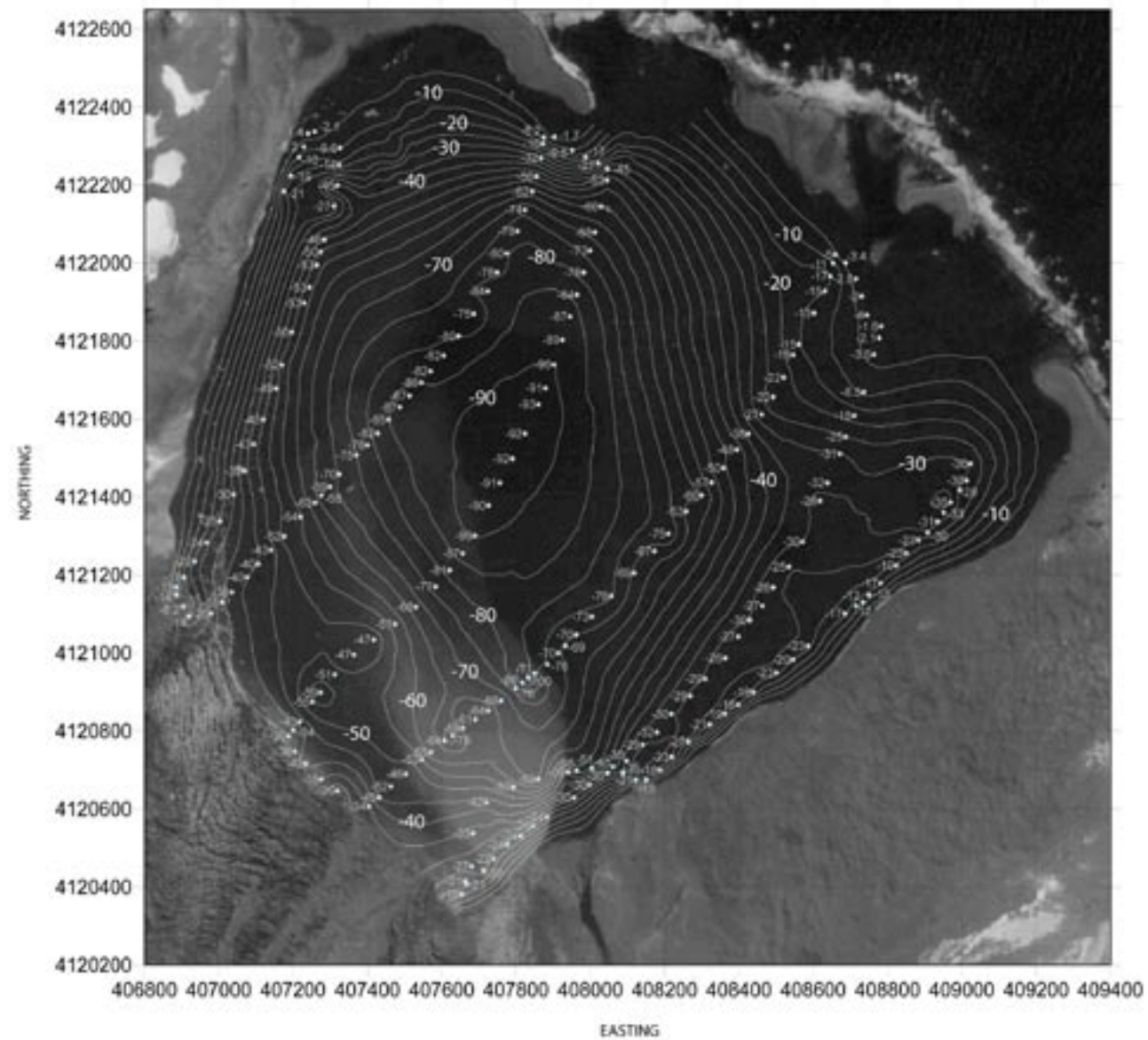


Figure A3.4 The shrinking Stephenson Glacier, the growing Stephenson Lagoon: a photographic record from Scarlet Hill. a) February 19th 1988. Photo F61/4 © Jenny Scott. b) November 29th 2000. c) February 5th 2004.

Figure A3.5 Bathymetry of Compton Lagoon.



Bathymetric survey 3 Feb 2004 by M. Truffer, S. Donoghue, C. Stephenson.
 Depths relative to sea level.
 Positional error ± 5 m. Depth error: ± 1 m where shallower than -30 m;
 $\pm 5\%$ where deeper than -30 m.
 Horizontal datum WGS84. Map projection UTM Zone 43
 Background image Quickbird Satellite panchromatic acquired 12 Jan 2003
 Map compilation by D. Thost, Australian Antarctic Division

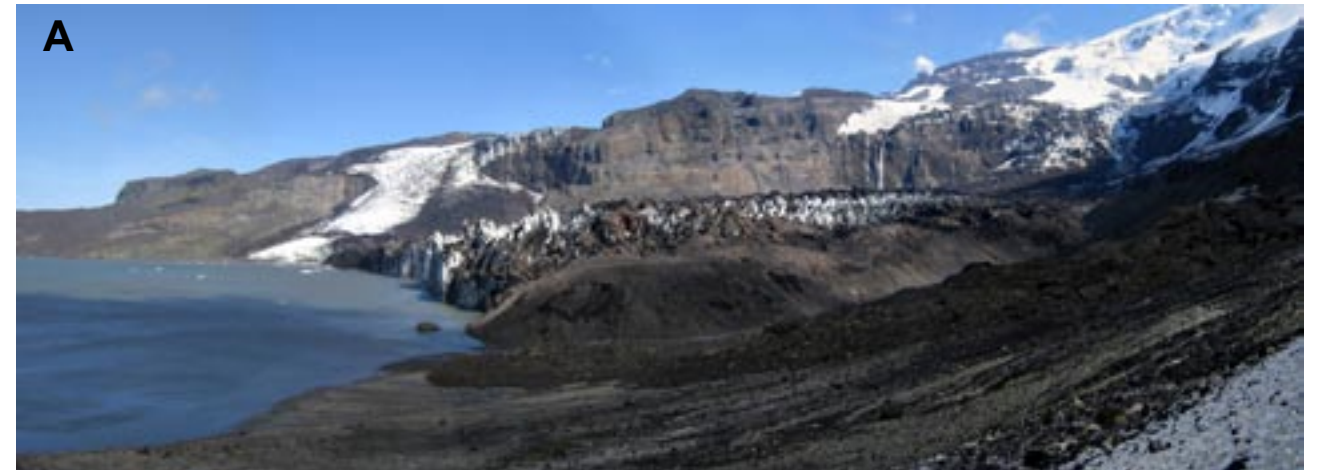


Figure A3.6 a) Panorama from the south-western corner of Compton Lagoon, showing the lagoon and Compton Glacier valley. Long Ridge is on the skyline to the right. February 3rd 2004.

b) The heavily seracated terminus of the Compton Glacier. The vertical ice front is estimated at ~30 m high. Note the recumbent fold in the ice (arrow points to the fold nose).



Figure A3.7 Bathymetry of Winston Lagoon.

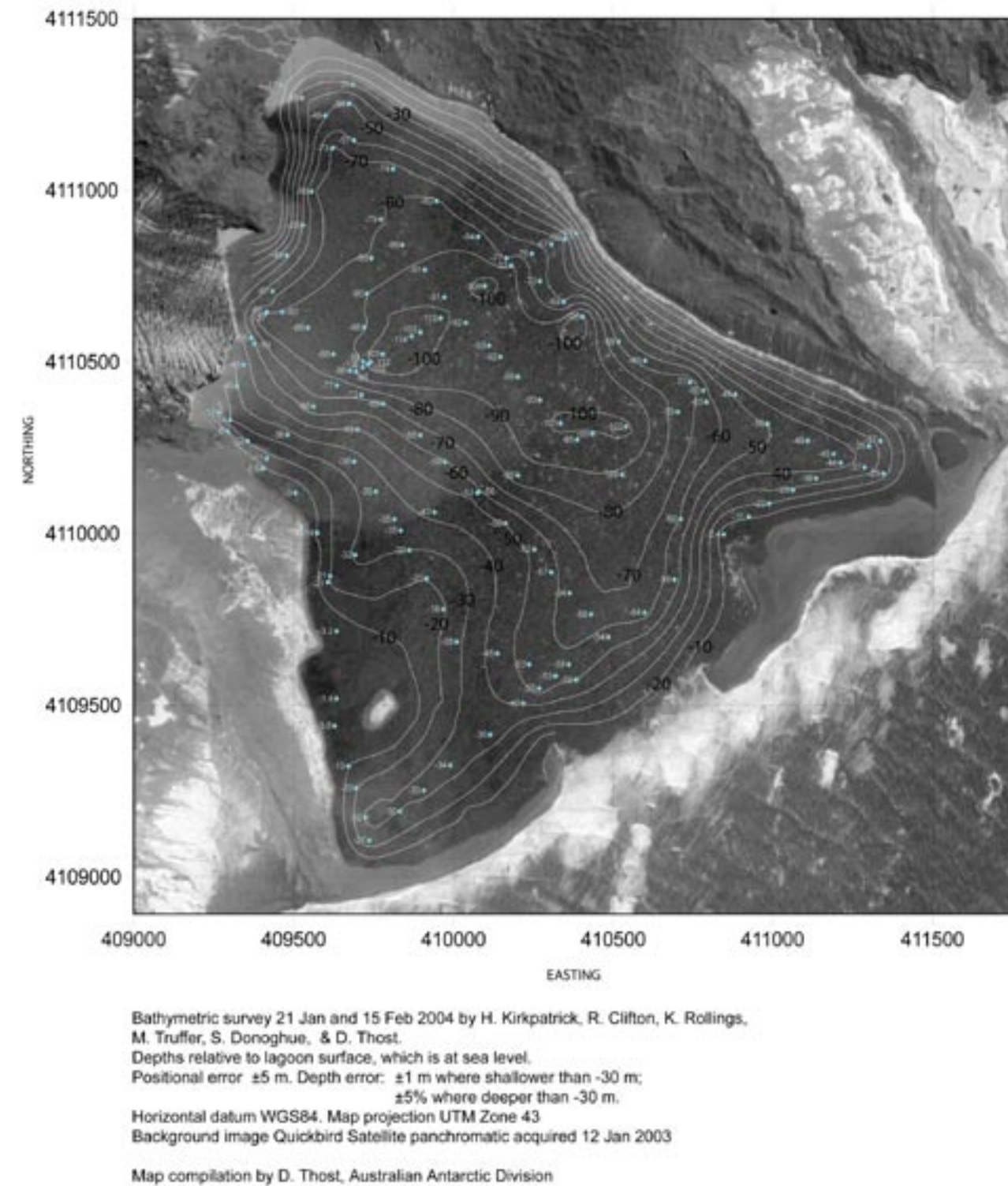
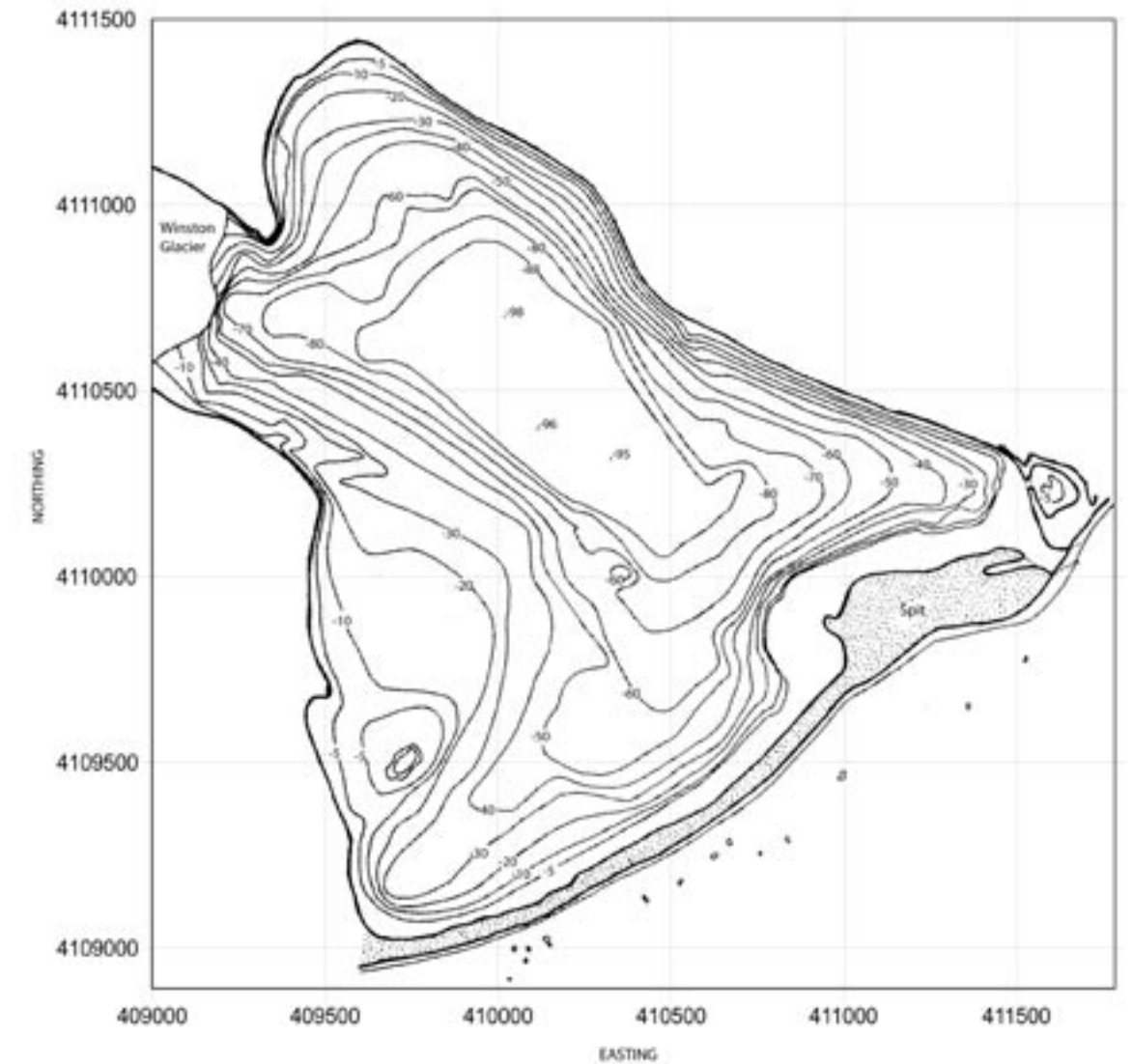


Figure A3.8 Panorama of Winston Lagoon looking southwest towards Cape Lockyer (centre).

Figure A3.9 Bathymetric map of Winston Lagoon by Kirkwood & Burton (1988), modified to align with the UTM grid. The bathymetry was acquired by Roger Kirkwood and Jenny Scott using a Furuno chart recorder echo sounder, from 38 transects across the lagoon. Note the increased detail in such features as the ridge paralleling the southern margin of Winston Glacier. Also noteworthy is the dynamic nature of the cobble/boulder spit which was breached between March 2000 and January 2003.



A4 Panorama photo record



Figure A4.1 360° panorama from the Brown Hut camp area.



Figure A4.2 360° panorama from the south-eastern lateral moraine of Brown Glacier. A Rock Hopper penguin colony is located in the group of rocks to the left.



Figure A4.3 360° panorama from the south-eastern end of the Brown Lagoon bar.

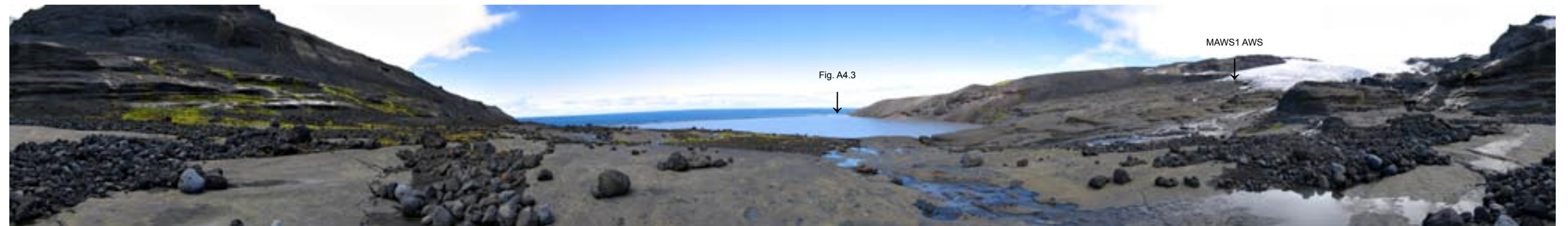


Figure A4.4 360° panorama from the glacial pavement beneath Round Hill.

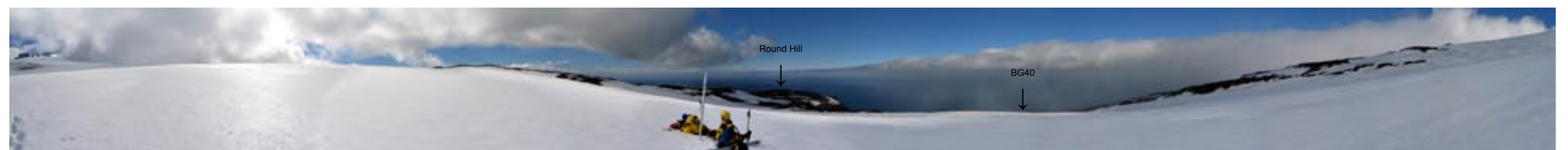


Figure A4.5 Near-360° panorama from BG35 survey point on Brown Glacier, on December 28th 2003.

Figure A4.6 360° panoramas from the 550 m AWS site adjacent to Brown Glacier. a) On December 27th 2003. b) On January 7th 2004.





Figure A4.7 Panorama from MAWS2 AWS site (920 m a.s.l.) adjacent to Brown Glacier (far left). Elephant Spit is visible to the east in the centre of the photograph. January 16th 2004.

Figure A4.8 Panorama from MAWS2 AWS site, with Long Ridge in the centre of the photograph. Brown Glacier flows out of frame to the right. January 16th 2004.

